A metaplectic Jacquet–Langlands transfer for division algebras of prime degree, II: matching of arbitrary functions and an improved global transfer

Draft 9

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Abstract

This is the second part of a study of a metaplectic Jacquet–Langlands transfer between the n-fold cover of $D_{\mathbb{A}}^{\times}$ constructed in [JC] and the Kazhdan–Patterson cover of $\mathrm{GL}(r, \mathbb{A}_K)$, where D is a central division algebra of prime degree r over a totally imaginary number field K containing μ_n, n odd and $(n, r) = 1$. Part I proved matching of test functions supported on the regular set, and a transfer of packets under a non-cancellation hypothesis. Here we remove the regular-support condition. Near the center, an equivariant section of the cover, built from canonical n-th roots, identifies genuine functions on either cover with ordinary functions on the underlying linear group, with the same orbital integrals, compatibly with the identification of elliptic tori of Part I (*Theorem D*). Combined with Part I away from the center, this gives matching of arbitrary genuine test functions on the cover of $D_v^{\times}$, and of all genuine test functions with vanishing non-elliptic orbital integrals on the cover of $\mathrm{GL}(r, K_v)$ (*Theorem E*). Beyond the results imported from Part I and the standard structure theory of the covers, the new external inputs concern the linear groups: orbital integrals of the Steinberg pseudo-coefficient and the linear transfer of cuspidal functions, which we derive from results of Flicker–Kazhdan and Badulescu. As an application we prove *Theorem C′*, an improvement of Theorem C of Part I: the packet identity holds with arbitrary matching test functions at all but one ramified place, and the non-cancellation hypothesis is replaced by a weaker one.

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1 Introduction

This section states the main results, explains how they improve the results of Part I, fixes conventions and labels, and lists what is left after this paper, with what Part III has since done. It depends only on the statements of Part I, and its notation is that of Part I; all notation is collected in §8, and the results are listed with their labels in the Summary of results that follows this section. References such as "Part I, Theorem 4.5" are to Draft 10 of Part I, whose numbering is that of Draft 8, and [JC] refers to the corrected version of [J] (`metaplectic_kazhdan_revised12.tex`).

1.1 Setting

We use the global data of Part I without change.

The standing assumptions are those of Part I, Assumption 1.1: K is a totally imaginary number field containing μ_n, with n odd; L/K is cyclic of prime degree r with $(n, r) = 1$; D is a central division algebra over K of dimension r^2 containing L; S is the set of places where D ramifies; and $c_v = r'c$ with $n \mid r - 1 + 2rc_v$. Both the cover $\overline{G}_D(\mathbb{A}_K)$ of $D^\times_{\mathbb{A}}$ from [JC, §6] and the Kazhdan–Patterson cover $\overline{G}(\mathbb{A}_K)$ of $\mathrm{GL}(r, \mathbb{A}_K)$ with parameter c_v are subgroups of one cover of $\mathrm{GL}(r, \mathbb{A}_L)$ (Part I, §§2 and 5). We use the splitting conventions of Part I, Lemma 5.1, good elements, the canonical identification $\jmath$ of covered elliptic tori on good elements (Part I, Theorem 3.9 and Corollary 3.12), the normalized orbital integrals $F = \Delta\Phi$ (Part I, Definition 4.1), and matching with the sign $(-1)^{r-1}$ (Part I, Definition 4.3).

1.2 Main results

We state the three main results informally, with pointers to the precise statements.

Part I left open the matching of test functions that do not vanish near the center: by Part I, Proposition 4.6, the orbital integrals of a genuine function $\tilde{f}_D$ on $\overline{G}_D(K_v)$ near the center are a nonzero constant times Δ, so a matching function on $\overline{G}_v$ cannot be supported in the regular set. The key new tool is the following.

Theorem D (Propositions 3.3 and 3.6)**.** *Let v be a finite place. On a conjugation-invariant open and closed neighborhood V of the unipotent set, the map $g \mapsto \sigma(g) := \widetilde{g^{1/n}}^{\,n}$ (canonical n-th root, any lift, n-th power) is a well-defined continuous section of the cover. It is equivariant under conjugation, consists of good elements at regular points, agrees with the splittings over small open subgroups and over unipotent radicals, and is compatible with $\jmath$. Genuine functions supported on $p^{-1}(ZV)$ correspond bijectively to ordinary functions on G/Z supported on ZV, with the same orbital integrals, on both covers.*

Combined with Part I away from the center, Theorem D gives:

Theorem E (Theorem 5.8)**.** *Let $v \in S$. Every genuine test function on $\overline{G}_D(K_v)$ has a matching genuine test function on $\overline{G}_v$. Every genuine test function on $\overline{G}_v$ whose orbital integrals vanish at good regular non-elliptic elements has a matching function on $\overline{G}_D(K_v)$.*

The matching function in the first part of Theorem E is explicit: $\tilde{f} = \tilde{f}_D(1)\,\tilde{f}^{\mathrm{St}} + \tilde{f}_{\mathrm{reg}}$, where $\tilde{f}^{\mathrm{St}}$ is the transport of the linear Steinberg pseudo-coefficient (§5.2) and $\tilde{f}_{\mathrm{reg}}$ is supported on the good regular elliptic set. We also determine the germ near the center that any matching function must have (Theorem 5.13).

Three objects should be kept apart: the Steinberg pseudo-coefficient f_{St} on $\mathrm{GL}(r, F)$; the genuine function $\tilde{f}^{\mathrm{St}}$ on $\overline{G}_v$, its transport near the center (cut off to a neighborhood of $\overline{Z}_v$ and rescaled); and pseudo-coefficients of genuine irreducible representations of $\overline{G}_v$. The second is *not* an instance of the third (Remark 5.6); matching needs only its orbital integrals.

Theorem C′ (Theorems 6.5 and 6.7)**.** *Let $\overline{\pi}_D$ be a genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ with a supercuspidal component at a finite place $w_0 \notin S$. The packet identity of Part I holds with arbitrary matching test functions at all places of S except one, u, which may be chosen freely. If the D-side packet satisfies the non-cancellation hypothesis* (H'_D) *(Definition 6.4), which is implied by the hypothesis* (H_D) *of Part I, then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with $\overline{\pi}_v \cong \overline{\pi}_{D,v}$ for all $v \notin S$. Conversely, if $\overline{\pi}$ is the only member of its* GL*-side packet and its components at the places of S are square-integrable, then $\overline{\pi}$ transfers to $\overline{G}_D(\mathbb{A}_K)$.*

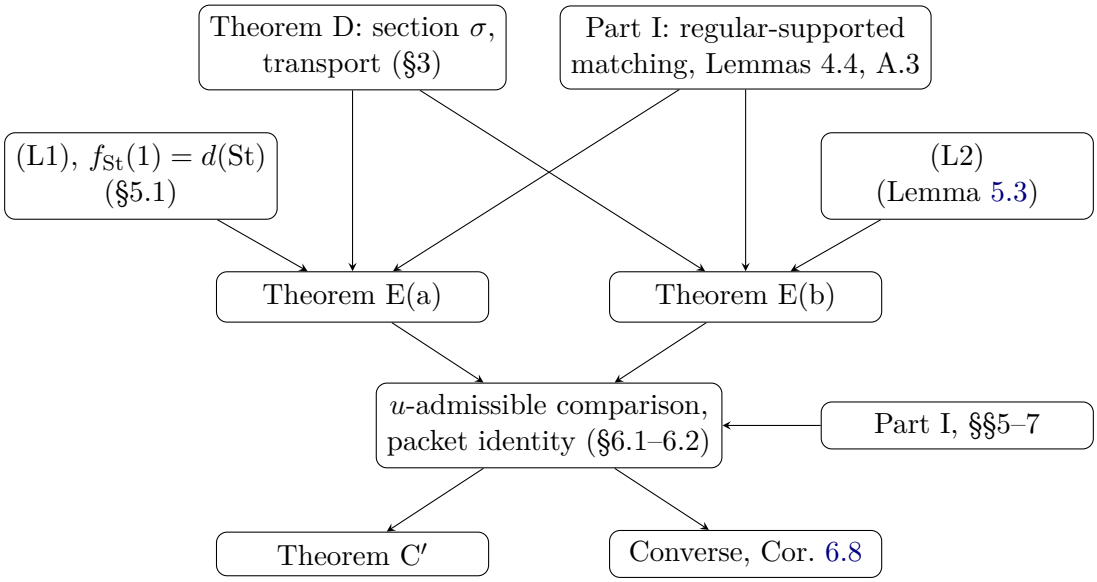
1.3 What changes from Part I

We present every new global result as an improvement of a numbered result of Part I. The following table is a guide for the reader of both papers; each result in Part II that improves a result of Part I ends with a paragraph headed "What changes from Part I".

Part I	Part II	New input
Thm. 4.5 (Theorem B: regular-supported matching)	Thm. 5.8 (Theorem E: arbitrary functions)	Theorem D; (L1), (L2)
Prop. 4.6 (orbital integrals near $\overline{Z}_v$, D side)	Thm. 5.13 (required germ, GL side)	Theorem D; measures
Lemma A.3 (regular support)	Lemma 4.2 (arbitrary $\tilde{f}$)	Lemma 4.1
Def. 6.5, Thm. 6.6 (comparison)	Def. 6.1, Thm. 6.2	Theorem E
Prop. 7.6 (packet identity)	Prop. 6.3	Theorem E
Def. 7.9, Thm. 7.11 (Theorem C, (H_D))	Def. 6.4, Thm. 6.5 (Theorem C′, (H'_D))	Prop. 6.3
Thm. 7.12, Rem. 7.13 (converse, (H))	Thm. 6.7, Cor. 6.8	Theorem E(b); pseudo-coefficients
Lemma 7.8 (cancellation)	Prop. 6.9	choice of u
§8, items (D1)–(D4)	§7	–

1.4 Logical structure

The dependencies among the main results are as follows; arrows mean "is used in".



The germ results (Corollary 3.7, Lemma 5.10, Corollary 5.11, Theorem 5.13(b)) and the cancellation criterion (Proposition 6.9) are not used in the proofs of Theorems D, E and C′.

1.5 Conventions and labels

We keep the conventions and the labeling scheme of Part I. They are collected, with the rest of the series' notation and the cocycle normalization, in the shared glossary of §8 (labels in §8.1, covers and the relabeling of μ_n in §8.5, Haar measures in §8.8).

All Hilbert symbols are n-th Hilbert symbols, and μ_n is relabeled once by $\zeta \mapsto \zeta^{r'}$ as in Part I, §1.4. Statements are marked by status:

- **[P]** *Proved here*: a complete argument is included, possibly using results marked **[E]**. For global statements, "proved" means proved relative to the cited results of Part I, which are restated in §2.3.
- **[E]** *Established elsewhere*: a precisely cited theorem, used as a black box. Several of these are results of Kazhdan [Kaz] that Flicker and Kazhdan state for covers in [FK]; we cite them in the form stated in [FK].
- **[O]** *Proof outlined*: the steps are identified, but at least one remains to be written out or checked.
- **[C]** *Conjectural*: requires new mathematical input (later parts of the series).

Statement	Where	Status
Compatible measures at S; Part I unaffected	§2.2	[**P**]
n-th roots; the section σ; compatibility with $\jmath$ (Theorem D)	§§3.1–3.2	[**P**]
Transport; regular elements near $\overline{Z}_v$ are good (Theorem D)	§3.3	[**P**]
Germ expansion at central good elements	Cor. 3.7	[**P**] using [**E**]
Orbital integrals of arbitrary genuine functions (r prime)	§4	[**P**]
Linear inputs (L1), (L2), $f_{\mathrm{St}}(1) = d(\mathrm{St})$	§5.1	[**P**] using [**E**]
The pair $(e_D, \tilde{f}^{\mathrm{St}})$	Prop. 5.5	[**P**]
Matching of arbitrary functions (Theorem E)	Thm. 5.8	[**P**]
Required germ near the center: orbital-integral form	Thm. 5.13(a)	[**P**]
Unipotent orbital integrals of functions cuspidal near 1; the trivial germ	Lemma 5.10, Cor. 5.11	[**P**] using [**E**]
Required germ: unipotent form (G1), (G2)	Thm. 5.13(b)	[**P**]
Comparison and packet identity with u-admissible pairs	§§6.1–6.2	[**P**] relative to Part I
Theorem C′ and its converse	§6.3	[**P**] relative to Part I, under (H'_D), (H′)
Cancellation criterion for (H'_D); finitely many S-types suffice	Prop. 6.9, Cor. 6.11	[**P**]
Local character relations; unconditional transfer	§7	[**C**] (character relations at tame places: Part III)

The running example

The series' running example is $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4 - \zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$, with $\mathfrak{p}_3 = (1 - \zeta_3)$ and $\mathfrak{p}_7 = (3 + \zeta_3)$ (Part I, §1, and `notes/mjl_running_example.tex`). Definition 2.1 fixes $\mathrm{vol}(D_v^\times / F^\times) = d(\mathrm{St})^{-1}$ at both places of S. Theorem E applies at both. At the wild place $\mathfrak{p}_3$, where the covered elliptic tori are non-abelian, matching near the center uses Theorem D. In Theorem C′ the place u is $\mathfrak{p}_3$ or $\mathfrak{p}_7$, and (H'_D) is a condition on test functions at these two places.

1.6 Organization

Section 2 recalls what we use from Part I and fixes the Haar measures at the ramified places. Sections 3–5 contain the local theory: the section σ and the transport (§3), orbital integrals of arbitrary genuine functions away from the center (§4), and matching of arbitrary functions (§5). Section 6 proves Theorem C′. Section 7 lists what is left for

Part III. Section 8 collects the notation of the series and the notation specific to Part II, and Section 9 discusses the work related to this Part; the literature of the whole series is reviewed in Part I, §11.

Summary of results

The results of Part II, with status labels (§8.1) and pointers; notation as in §2 and §8. "Relative to Part I" means that the cited results of Part I, restated in §2.3, are used.

1. **Compatible measures** (Definition 2.1, Remark 2.2; [**P**]). At $v \in S$, $\mathrm{vol}(D_v^\times/F^\times) = d(\mathrm{St})^{-1}$; every result of Part I holds unchanged with these measures.

2. **Theorem D: the section σ and transport** (Lemma 3.1, Propositions 3.3 and 3.6; [**P**]). At every finite place, including those above n, and on both covers: a canonical n-th root on a conjugation-invariant neighborhood V_M of the unipotent set gives a section $\sigma(g) = \widetilde{g^{1/n}}^{\,n}$ that is equivariant, good at regular points, compatible with the splittings and with $\jmath$. The map $\mathcal{T}$ identifies genuine functions supported on $p^{-1}(ZV_M)$ with functions on G/Z supported on ZV_M, preserving orbital integrals.

3. **Germs at central good elements** (Corollary 3.7; [**P**] using [**E**]). They are the Shalika germs of $\mathrm{GL}(r,F)$; on the D side only the trivial germ occurs.

4. **Orbital integrals of arbitrary genuine functions** (Lemmas 4.1 and 4.2; [**P**]). Away from $\overline{Z}$ they are admissible in the sense of Part I, §4.3 (r prime).

5. **Linear inputs and the Steinberg-type pair** (§5.1, Proposition 5.5). (L1) is [**E**]; (L2) is [**P**] using [**E**] (Lemma 5.3); $f_{\mathrm{St}}(1) = d(\mathrm{St})$ (Lemma 5.2; [**P**]). The genuine pair $(e_D, \tilde{f}^{\mathrm{St}})$ matches ([**P**]), but $\tilde{f}^{\mathrm{St}}$ is not a pseudo-coefficient of a single genuine representation (Remark 5.6).

6. **Theorem E: matching of arbitrary functions** (Theorem 5.8; [**P**] using (L1), (L2)). Every genuine $\tilde{f}_D$ has a matching $\tilde{f} = \tilde{f}_D(1)\tilde{f}^{\mathrm{St}} + \tilde{f}_{\mathrm{reg}}$, and every genuine $\tilde{f}$ with vanishing non-elliptic orbital integrals has a matching $\tilde{f}_D$.

7. **The required germ** (Theorem 5.13; [**P**]; Lemma 5.10 and Corollary 5.11, [**P**] using [**E**]). Near $\overline{Z}_v$ a function matches $\tilde{f}_D$ exactly when its non-trivial unipotent orbital integrals vanish and $\tilde{f}(1) = \tilde{f}_D(1)$.

8. **Theorem C′** (Theorem 6.2, Proposition 6.3; [**P**] relative to Part I; Theorem 6.5, [**P**] under (H'_D)). The packet identity holds with arbitrary matching functions at all places of S but one, and $\overline{\pi}_D$ transfers under (H'_D), which is implied by (H_D). The converse holds under (H′) (Theorem 6.7), in particular for a single-member packet with square-integrable components at S (Corollary 6.8; [**P**] using [**E**]).

9. **When (H'_D) fails** (Proposition 6.9, Corollary 6.11; [**P**]). Exactly when the S-multiplicities are proportional to the Plancherel weights; it holds if the levels at S are bounded.

10. **What is left** (§7; [**C**]). Local character relations (since proved by Part III at tame places in the range of the corrected Flicker–Kazhdan correspondence), an unconditional global correspondence, removal of (H'_D) and (H'), and completing [JC, Remark 10.3].

Running example: Theorem E applies at both $\mathfrak{p}_3$ and $\mathfrak{p}_7$; at the wild place $\mathfrak{p}_3$ matching near the center uses Theorem D.

2 Recollections from Part I, and Haar measures

This section recalls the notation of Part I that is used throughout, and fixes the Haar measures at the ramified places once and for all. It depends only on Part I, whose results used here are restated in §2.3 as (I1)–(I11). It introduces the local notation P_g, A_g, Φ^z, the compatible measures of Definition 2.1, and the constant λ_v.

2.1 Local notation

We fix the local notation used in §§3–5.

Throughout §§3–5, v is a finite place, $F = K_v$, $\mathfrak{p}$ its maximal ideal, ϖ a uniformizer and p the residual characteristic. We write G for either $\mathrm{GL}(r, F)$ (embedded in $\mathcal{G}_v$ by ι_{GL}) or $D_v^\times$ (embedded by ι_D), $\overline{G} = p^{-1}(G) \subset \overline{\mathcal{G}}_v$, $Z = F^\times$ and $\overline{Z} = p^{-1}(Z)$, which is central and common to both covers (Part I, Lemma 2.2). For $g \in G$ let P_g be its characteristic polynomial (reduced characteristic polynomial if $G = D_v^\times$), $A_g := F[g]$ the commutative F-algebra it generates, and call the roots of P_g in $\overline{F}$ the eigenvalues of g. For a genuine character $\tilde{\theta}$ of $\overline{Z}$, $C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ is the space of smooth genuine functions with $\tilde{f}(z\tilde{g}) = \tilde{\theta}(z)^{-1}\tilde{f}(\tilde{g})$ and support compact modulo $\overline{Z}$. Orbital integrals Φ and $F = \Delta\Phi$ are those of Part I, Definition 4.1, with $\Delta(\gamma) = |D(\gamma)|^{1/2}|\det\gamma|^{-(r-1)/2}$; as in [FK, §6], $D(\gamma)$ is formed from the distinct eigenvalues. We also use the variant $\Phi^z(\gamma, f) = \int_{Z\backslash G} f(g^{-1}\gamma g)\,dg$ of [FK, §6] for regular elliptic γ; then $\Phi^z(\gamma, f) = \mathrm{vol}(T_\gamma/Z)\,\Phi(\gamma, f)$.

2.2 Haar measures at the ramified places

We fix compatible Haar measures at the places of S (Definition 2.1), used in the rest of the paper. This choice is made once and for all; it simplifies the normalization at the identity and the formal-degree relations of §7(ii).

Measures are as in Part I, §4.1: dg on G/Z, dt on T/Z, and dg/dt on $T\backslash G$; at $v \in S$ they are specialized as follows.

Definition 2.1 (Compatible measures)**.** At each $v \in S$, let dg_v be any Haar measure on $\mathrm{GL}(r, F)/F^\times$, and let $d(\mathrm{St}) = d(\mathrm{St}; dg_v)$ be the formal degree, with respect to dg_v, of the Steinberg representation of $\mathrm{GL}(r, F)$ (trivial central character). Normalize the Haar measure $dg_{D,v}$ on the compact group $D_v^\times/F^\times$ by

$$\mathrm{vol}(D_v^\times/F^\times;\, dg_{D,v}) = d(\mathrm{St}; dg_v)^{-1}.$$

The measures dt on tori are arbitrary, as in Part I (matching does not depend on them).

Remark 2.2. (1) [**P**] Part I, §5.3, allows arbitrary dg_v and $dg_{D,v}$ at $v \in S$, provided matching and the trace formulas use the same ones. So every result of Part I, local and global, holds with the measures of Definition 2.1, and we cite Part I without change.

(2) The formal degree of the trivial representation of $D_v^\times$ is $\mathrm{vol}(D_v^\times/F^\times)^{-1}$, and the trivial representation corresponds to St under the linear Jacquet–Langlands correspondence [Bad, Theorem 2.2]. So Definition 2.1 says that this pair has equal formal degrees. We expect that it agrees with Kottwitz's normalization of measures on inner forms [Kot88], but nothing below uses this.

(3) For arbitrary measures, the constant $\lambda_v := d(\mathrm{St})\,\mathrm{vol}(D_v^\times/F^\times)$ would appear in the normalization at the identity (Theorem 5.13(c)). Definition 2.1 is exactly the condition $\lambda_v = 1$.

2.3 Statements used from Part I

For the reader's convenience, and to make the dependence on Part I auditable, we restate the results of Part I that are used, in the form used. Numbers refer to Part I, Draft 10, whose numbering is that of Draft 8. The labels (I1)–(I11) are used in the rest of the paper.

(I1) *Admissible functions* (§4.3). For $\bullet \in \{D, \mathrm{GL}\}$ and an elliptic torus of type i, a function h on the good regular elements $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ is admissible if it is (i) locally constant, (ii) satisfies $h(z\tilde{t}) = \tilde{\theta}_v(z)^{-1}h(\tilde{t})$ for $z \in \overline{Z}_v$, (iii) has support compact modulo $\overline{Z}_v$ with closure contained in the regular set, and (iv) is invariant under the normalizer of the torus.

(I2) *Lemma 4.4.* (b) $h \mapsto h \circ \jmath_i^{-1}$ is a bijection between admissible functions on the two sides. (c) Every admissible h equals $F(\cdot, \tilde{f}_i)$ on $Z(\overline{E}_{i,\bullet})^{\mathrm{reg}}$ for some $\tilde{f}_i \in C_c^\infty$ supported on the good regular elements conjugate into that torus; $F(\tilde{\delta}, \tilde{f}_i) = 0$ for regular $\tilde{\delta}$ not conjugate into it.

(I3) *Theorem 4.5.* (a) Every $\tilde{f}_D$ vanishing near $\overline{Z}_v$ has a matching $\tilde{f}$ supported on the good regular elliptic set. (b) Every $\tilde{f}$ supported in the regular set with vanishing orbital integrals at good regular non-elliptic elements has a matching $\tilde{f}_D$ vanishing near $\overline{Z}_v$.

(I4) *Proposition 4.6.* For $\tilde{f}_D$ there are a normal open compact $U \subset D_v^\times$ and a splitting κ_U with $F(z\kappa_U(t), \tilde{f}_D) = \Delta(t)\,\mathrm{vol}(T\backslash D_v^\times)\tilde{f}_D(z)$ for $z \in \overline{Z}_v$ and regular $t \in U$.

(I5) *Lemma A.1(b)* (Harish-Chandra). For $C \subset T^{\mathrm{reg}}$ and $\Omega \subset G$ compact modulo Z, the set of $\dot{x} \in T\backslash G$ with $x^{-1}Cx \cap \Omega \neq \emptyset$ is compact. (Stated in Part I for any such Ω.)

(I6) *Lemma A.3.* For regular-supported $\tilde{f}$, $F(\cdot, \tilde{f})$ on an elliptic torus is admissible. *Lemma A.5.* Traces of regular-supported functions depend only on their good regular orbital integrals.

(I7) *Proposition 6.1.* The D-side trace formula: $\sum m(\overline{\pi}_D)\operatorname{tr}\overline{\pi}_D(\tilde{f}_D) = \operatorname{vol}\cdot\tilde{f}_D(1) + \sum_{[\gamma]} c(\gamma)F(\iota(\gamma),\tilde{f}_D)$, the sum over rational regular classes modulo the center that are good everywhere; every non-central rational element of $G_D(K)$ is regular since r is prime. No condition on the test function at S.

(I8) *Proposition 6.2* (variant of [FK, §18]). For $\tilde{f}$ with (i) a supercusp form at some finite place w_0 and (ii) support in the regular elliptic set at some finite place u, and no condition elsewhere: $\sum_{\text{cuspidal}} m(\overline{\pi})\operatorname{tr}\overline{\pi}(\tilde{f}) = \sum_{[\gamma]} c(\gamma)F(\iota(\gamma),\tilde{f})$, the sum over rational regular elliptic classes good everywhere.

(I9) *Theorem 5.6 and Corollary 5.7.* If $\tilde{f}_{D,v}$ and $\tilde{f}_v$ match at every $v \in S$ (no support condition) and correspond via ψ_v outside S, then $F(\iota(\gamma),\tilde{f}_D) = F(\iota(j\gamma),\tilde{f})$, the local signs multiplying to $(-1)^{(r-1)|S|} = 1$ (if r is odd, $r-1$ is even; if $r=2$, $|S|$ is even because the local invariants of D sum to 0 and K has no real places); and classes whose field does not embed in D contribute 0 whenever (M2) holds at all $v \in S$, since such a class is non-elliptic at some $v \in S$ (Albert–Brauer–Hasse–Noether).

(I10) *Lemma 7.2.* For fixed levels at the finite places (with any test functions bi-invariant under them) and fixed K_∞-types and infinitesimal character, only finitely many automorphic representations contribute on either side.

(I11) *Lemma 7.8 (proof).* For a level U_v at $v \in S$ (split by κ_v, with $\tilde{\theta}_v \circ \kappa_v$ trivial on $U_v \cap K_v^\times$), the corner algebra satisfies $\mathcal{H}_v \cong \bigoplus_{\tau_v \in \mathcal{T}_v} \operatorname{End}(\tau_v^{U_v})$, and $\operatorname{ev}_{1,v} = \operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v)^{-1}\sum_{\tau_v}(\dim\tau_v)\operatorname{tr}\tau_v$ on $\mathcal{H}_v$; $\mathcal{R}_v = \ker\operatorname{ev}_{1,v}$.

3 Harmonic analysis near the center

This section proves Theorem D. We construct a canonical n-th root on a conjugation-invariant neighborhood of the unipotent set (§3.1), lift it to an equivariant section σ of the cover (§3.2), and use σ to transport genuine functions supported near $\overline{Z}$ to ordinary functions on G/Z, with the same orbital integrals (§3.3). As a corollary, the germ expansion of genuine orbital integrals at central good elements is the Shalika germ expansion of the linear group (§3.4). Everything in this section holds at every finite place, including those above n, and on both covers. The section depends on Part I, §§2–3 (the covers, scalar centrality and the identification $\jmath$) and on the Shalika germ expansion for $\mathrm{GL}(r,F)$ [Sha72]. It introduces the neighborhoods V_M, the constants δ_n, B, M_0, M_1, the root $g^{1/n}$, the section σ and the transport $\mathcal{T}$.

3.1 n-th roots near the unipotent set

We define a canonical n-th root on a conjugation-invariant neighborhood of the unipotent set, and introduce Ω_M, V_M, δ_n, B, M_0 and $g^{1/n}$, which are used throughout. The point of using n-th roots is that the n-th power of a lift does not depend on the lift.

For $M \geq 1$ let Ω_M be the set of monic polynomials of degree r over F that are congruent to $(X-1)^r$ coefficientwise modulo $\mathfrak{p}^M$, and put

$$V_M := \{g \in G : \ P_g \in \Omega_M\}.$$

Absolute values. In this subsection $q = |\mathcal{O}/\mathfrak{p}|$, $e = v_F(p)$ is the absolute ramification index, and $|x|_F = q^{-v_F(x)}$ is the normalized absolute value of F, extended uniquely to $\overline{F}$; on $\mathcal{O}$-algebras we use the corresponding sup norm $\|\cdot\|$ on coordinates. (In terms of the p-adic absolute value $|\cdot|_p$ with $|p|_p = p^{-1}$, one has $|x|_F = |x|_p^{[F:\mathbb{Q}_p]}$.) Put

$$\delta_n := \begin{cases} 1, & p \nmid n, \\ q^{-e/(p-1)}, & p \mid n, \end{cases} \qquad B := |n|_F^{-1} q^{e/(p-1)}.$$

Lemma 3.1. ***[P]***

(a) V_M is open, closed and stable under conjugation, and so is ZV_M. Every eigenvalue λ of $g \in V_M$ satisfies $|\lambda - 1|_F \leq |\varpi|_F^{M/r}$.

(b) If $u \in \overline{F}$, $u^n = 1$ and $|u-1|_F < \delta_n$, then $u = 1$.

(c) $|\binom{1/n}{k}|_F \leq B^k$ for all $k \geq 0$. There is M_0 such that for $M \geq M_0$ and $g \in V_M$ the series $g^{1/n} := \sum_{k\geq 0} \binom{1/n}{k}(g-1)^k$ converges in A_g. Its sum is the unique $h \in A_g$ with $h^n = g$ whose eigenvalues μ satisfy $|\mu - 1|_F < \delta_n$.

(d) For $M \geq M_0$: $(xgx^{-1})^{1/n} = xg^{1/n}x^{-1}$ for $x \in G$; the map $g \mapsto g^{1/n}$ is continuous on V_M; and $(ag)^{1/n} = a^{1/n}g^{1/n}$ whenever $a \in F^\times$ and $a, g, ag \in V_{M_0}$.

Proof. (a) The map $g \mapsto P_g$ is continuous and Ω_M is open and closed, so V_M is open and closed; it is conjugation-stable because P_g is. If $\mu = \lambda - 1$, then μ is a root of $P_g(X+1) = X^r + \sum_{i<r} c_i X^i$ with $|c_i|_F \leq |\varpi|_F^M$. If $|\mu|_F > 1$, then $|\mu|_F^r \leq |\varpi|_F^M |\mu|_F^{r-1}$ gives $|\mu|_F < 1$, a contradiction; so $|\mu|_F \leq 1$ and $|\mu|_F^r \leq |\varpi|_F^M$. For ZV_M: openness is clear. If $a_k g_k \to g$ with $g_k \in V_M$, then $\det g_k \in \mathcal{O}^\times$ (its reduced norm, when $G = D_v^\times$), so $|a_k|_F^r \to |\det g|_F$ and the a_k lie in a compact subset of $F^\times$; passing to a convergent subsequence $a_k \to a$, we get $g_k \to a^{-1}g \in V_M$, so $g \in ZV_M$.

(b) If $u \neq 1$, then u is a root of unity of order $m > 1$ dividing n. If $p \nmid m$, the reduction of u is a nontrivial root of unity in the residue field, so $|u-1|_F = 1$. If $p \mid m$, some power u^{m/p^a} is a primitive p^a-th root of unity ζ with $a \geq 1$, and $|u-1|_F \geq |\zeta - 1|_F \geq |\zeta_p - 1|_F = |p|_F^{1/(p-1)} = q^{-e/(p-1)}$. (The first inequality holds because $u^j - 1 = (u-1)(u^{j-1} + \cdots + 1)$ with the second factor integral; the second because $|\zeta - 1|_F = |p|_F^{1/(p^{a-1}(p-1))}$ for ζ of exact order p^a.)

(c) Since $\binom{1/n}{k} = n^{-k}\prod_{j<k}(1-jn)/k!$ and $v_p(k!) \leq k/(p-1)$, we have $|\binom{1/n}{k}|_F \leq |n|_F^{-k}|k!|_F^{-1} \leq |n|_F^{-k} q^{ek/(p-1)} = B^k$. Put $y = g - 1$. Since P_g has integral coefficients, y is integral over $\mathcal{O}$, so $\mathcal{O}[y] \subset A_g$ is a finitely generated, hence complete, $\mathcal{O}$-module. By Cayley–Hamilton $y^r = -\sum_{i<r} c_i y^i \in \varpi^M \mathcal{O}[y]$, so $y^{kr+i} \in \varpi^{kM}\mathcal{O}[y]$. So the series converges in $\mathcal{O}[y]$ as soon as $|\varpi|_F^M B^r < 1$. Then $h^n = g$, by the identity $(\sum_k \binom{1/n}{k} X^k)^n =$

$1+X$ in $\mathbb{Q}[[X]]$ and absolute convergence. Each F-algebra map $\chi : A_g \to \overline{F}$ is continuous, and $\chi(g)$ runs over the eigenvalues of g. Hence the eigenvalues of h are the numbers $b(\lambda-1)$, where $b(x) = \sum_k \binom{1/n}{k} x^k$ and $|\lambda - 1|_F \le |\varpi|_F^{M/r}$. Since $|b(x)-1|_F \le \max_{k\ge 1}(B|x|_F)^k$, enlarging M_0 gives $|b(\lambda-1)-1|_F < \delta_n$. For uniqueness, if h' is another such root, then $u := h'h^{-1} \in A_g$ satisfies $u^n = 1$, and each eigenvalue of u is within δ_n of 1 (ultrametric inequality, since the eigenvalues of h are units). By (b) all eigenvalues of u equal 1. Since $X^n - 1$ is separable, u is semisimple, so $u = 1$.

(d) Equivariance and multiplicativity follow from the uniqueness in (c): $xg^{1/n}x^{-1} \in A_{xgx^{-1}}$ is an n-th root of xgx^{-1} with the right eigenvalues, and $a^{1/n}g^{1/n} \in A_g = A_{ag}$ is an n-th root of ag whose eigenvalues are products of numbers within δ_n of 1. For continuity, let $g_0 \in V_M$. Suppose first that $G = \mathrm{GL}(r,F)$, or $G = D_v^\times$ with D_v split, identified with $\mathrm{GL}(r,F)$ by some $D_v \cong M_r(F)$. Then $\Lambda := \mathcal{O}[g_0]\mathcal{O}^r$ is a g_0-stable lattice (finitely generated since g_0 is integral), and g_0^{-1} is also integral since $\det g_0 \in \mathcal{O}^\times$; so after conjugation $g_0 \in \mathrm{GL}(r,\mathcal{O})$. On the open set $V_M \cap \mathrm{GL}(r,\mathcal{O})$ we have $y \in M_r(\mathcal{O})$ and $\|y^r\| \le |\varpi|_F^M$, so the series converges uniformly there. (Note that $V_M \not\subset \mathrm{GL}(r,\mathcal{O})$: for instance $\left(\begin{smallmatrix} 1 & \varpi^{-N} \\ 0 & 1 \end{smallmatrix}\right) \in V_M$ for all M, N. This is why we conjugate.) If D_v is a division algebra, then $V_M \subset \mathcal{O}_D^\times$, since an element of D_v with integral reduced characteristic polynomial lies in the maximal order $\mathcal{O}_D$, and the same estimate holds in $\mathcal{O}_D$. Conjugation is a homeomorphism, so continuity holds everywhere. □

3.2 The section σ

We now lift the n-th root to the cover, defining the section σ (and σ_D, σ_{GL}) used in the rest of the paper. The resulting section is equivariant, consists of good elements, and is compatible with every splitting in sight and with the identification $\jmath$ of Part I.

Definition 3.2. For $M \ge M_0$ and $g \in V_M$ put $\sigma(g) := \tilde{h}^{\,n}$, where $h = g^{1/n}$ and $\tilde{h}$ is any lift of h to $\overline{G}$. We write σ_D and σ_{GL} when the group must be specified.

Since two lifts of h differ by $\zeta \in \mu_n$ and $\zeta^n = 1$, $\sigma(g)$ is well defined.

Proposition 3.3. ***[P]****Let $M \ge M_0$.*

(a) $\sigma : V_M \to \overline{G}$ is a continuous section of p, and $\sigma(xgx^{-1}) = \tilde{x}\,\sigma(g)\,\tilde{x}^{-1}$ for all $x \in G$ and any lift $\tilde{x}$.

(b) For every $g \in V_M$, $\sigma(g)$ commutes with $p^{-1}(\mathrm{Cent}_G(g))$. In particular, $\sigma(\gamma)$ is good for every regular $\gamma \in V_M$, and $\sigma(t) \in Z(\overline{T})$ for every $t \in T \cap V_M$ and every maximal torus T.

(c) If $a \in F^\times$ and $a, g, ag \in V_{M_0}$, then $\sigma(a) \in \overline{Z}$ and $\sigma(ag) = \sigma(a)\sigma(g)$.

(d) Let $W \subseteq V_{M_0}$ be an open compact subgroup over which p splits by a homomorphism κ. Then there is an open compact subgroup $U \subseteq W \cap V_M$ with $u^{1/n} \in W$ for all $u \in U$, and $\sigma|_U = \kappa|_U$. In particular, the splitting κ_U of Part I, Proposition 4.6, is $\sigma_D|_U$ (there $U = U_k$, $W = U_{k'}$, and $\kappa_U(t) = \kappa(s)^n$ with $s = t^{1/n} \in U_{k'}$).

(e) *If u is unipotent and lies in the unipotent radical N of a parabolic subgroup of $\mathrm{GL}(r,F)$, then $\sigma(u)=s_N(u)$, where s_N is the unique splitting of p over N.*

(f) (Compatibility of the two sides.) *For $a\in F^\times\cap V_{M_0}$, $\sigma_D(a)=\sigma_{\mathrm{GL}}(a)$ in $\overline{Z}$. Let $E\subset D_v$ be a maximal subfield, $j:E\hookrightarrow M_r(F)$ an embedding, and $\jmath_{E,j}:Z(\overline{E}_D)\to Z(\overline{E}_{\mathrm{GL}})$ the canonical identification of Part I, Theorem 3.9. Then $\jmath_{E,j}(z\,\sigma_D(t))=z\,\sigma_{\mathrm{GL}}(j(t))$ for $z\in\overline{Z}$ and $t\in E^\times\cap V_M$.*

Proof. (a) Near any point, $g\mapsto g^{1/n}$ is continuous (Lemma 3.1(d)) and p has continuous local sections, so $\tilde{h}$ may be chosen to depend continuously on g locally. Equivariance: $\tilde{x}\tilde{h}\tilde{x}^{-1}$ is a lift of $xhx^{-1}=(xgx^{-1})^{1/n}$, so $\tilde{x}\tilde{h}^n\tilde{x}^{-1}=(\tilde{x}\tilde{h}\tilde{x}^{-1})^n=\sigma(xgx^{-1})$.

(b) If $c\in\mathrm{Cent}_G(g)$, then c commutes with $A_g\ni h$, so for a lift $\tilde{c}$ we have $\tilde{c}\tilde{h}\tilde{c}^{-1}=\zeta\tilde{h}$ with $\zeta\in\mu_n$, and therefore $\tilde{c}\tilde{h}^n\tilde{c}^{-1}=\zeta^n\tilde{h}^n=\tilde{h}^n$.

(c) $a^{1/n}\in F^\times$, so its lifts lie in $\overline{Z}$, which is central. Hence $\sigma(ag)=(\widetilde{a^{1/n}}\,\tilde{h})^n=\widetilde{a^{1/n}}^n\tilde{h}^n=\sigma(a)\sigma(g)$, using Lemma 3.1(d).

(d) The root map $u\mapsto u^{1/n}$ is continuous at 1 and fixes 1, so some open compact subgroup $U\subseteq W\cap V_M$ satisfies $u^{1/n}\in W$ for $u\in U$. For such u, $\kappa(u^{1/n})$ is a lift of $u^{1/n}$, so $\sigma(u)=\kappa(u^{1/n})^n=\kappa(u)$, since κ is a homomorphism. We do *not* assume $u^{1/n}\in U$: when $p\mid n$ this generally fails for the standard filtration subgroups, since a p-th root of an element of $1+\mathfrak{p}^m$ lies only in a larger principal-unit group.

(e) $u^{1/n}=\sum_k\binom{1/n}{k}(u-1)^k$ is a finite sum, and it lies in N. Since s_N is a homomorphism, $\sigma(u)=s_N(u^{1/n})^n=s_N(u)$.

(f) The first assertion holds because $\iota_D(a)=\iota_{\mathrm{GL}}(a)$ and $a^{1/n}\in F^\times$, so both sides are the n-th power of a lift of the same element of $\mathcal{G}_v$. For the second, put $s=t^{1/n}\in F[t]\subseteq E$. Then $j(s)=j(t)^{1/n}$, by uniqueness in Lemma 3.1(c), since j is an F-algebra embedding and preserves eigenvalues. Let $\varphi\in\mathrm{Iso}_{\overline{Z}}(\overline{E}_D,\overline{E}_{\mathrm{GL}})$. It is a group isomorphism over $\iota_{\mathrm{GL}}j\iota_D^{-1}$, so $\varphi(\sigma_D(t))=\varphi(\tilde{s})^n$, and $\varphi(\tilde{s})$ is a lift of $j(s)$. Hence $\varphi(\sigma_D(t))=\sigma_{\mathrm{GL}}(j(t))$. By (b), $\sigma_D(t)\in Z(\overline{E}_D)$, where φ restricts to $\jmath_{E,j}$; and φ is the identity on $\overline{Z}$. □

Remark 3.4. (1) Proposition 3.3 holds equally for $v\mid n$; only M_0 grows. So the distinction between $v\mid n$ and $v\nmid n$, which governs the structure of the covered tori (Part I, Proposition 3.5), plays no role near the center.

(2) Part (e) says that σ extends the canonical lifting of unipotent elements. So the unipotent orbital integrals $\Phi(z\,\sigma(u),\tilde{f}):=\int_{G_u\backslash G}\tilde{f}(\tilde{x}^{-1}z\sigma(u)\tilde{x})\,d\dot{x}$ are well defined by (b), and they are the unipotent orbital integrals of [FK, §8], where unipotent elements are lifted by the splitting s of [FK, §2] (see Remark 3.8(1) for the normalization of F).

3.3 Transport to the linear group

We show that genuine functions supported in $p^{-1}(ZV_M)$ are the same thing as linear functions on G/Z supported in ZV_M, with the same orbital integrals. This holds on both sides and is compatible with $\jmath$. The constant M_1 and the transport $\mathcal{T}$ (and $\mathcal{T}_D$) defined here are used in §5.

Lemma 3.5 (Choice of M)**.** ***[P]****Let $\tilde{\theta}$ be a genuine character of $\overline{Z}$. There is $M_1\geq M_0$ such that for $M\geq M_1$:*

(a) if $a \in F^\times$, $g \in V_M$ and $ag \in V_M$, then $a \in 1 + \mathfrak{p}^{\lceil M/r \rceil} \subseteq V_{M_0}$;

(b) $\tilde{\theta}(\sigma(a)) = 1$ for all $a \in 1 + \mathfrak{p}^{\lceil M/r \rceil}$.

Proof. (a) If λ is an eigenvalue of g, then $a\lambda$ is one of ag, and Lemma 3.1(a) gives $|a-1|_F = |a\lambda - \lambda|_F/|\lambda|_F \le |\varpi|_F^{M/r}$. A scalar a with $|a-1|_F \le |\varpi|_F^m$ has $P_a = (X-a)^r \in \Omega_m$. (b) By Proposition 3.3(c), $a \mapsto \tilde{\theta}(\sigma(a))$ is a continuous character of $1+\mathfrak{p}^{M_0}$, so it is trivial on $1+\mathfrak{p}^m$ for m large. □

Measure convention. For $g \in G$ let G_g be its centralizer. If $g \in V_M$, then $\sigma(g)$ commutes with $p^{-1}(G_g)$ (Proposition 3.3(b)), so the genuine orbital integral $\Phi(z\sigma(g), \tilde{f}) = \int \tilde{f}(\tilde{x}^{-1} z\sigma(g)\tilde{x})\, d\dot{x}$ is an integral over the same quotient space $G_g\backslash G$ as the linear orbital integral at $p(z)g$. Throughout, the two are computed with the *same* quotient measure on $G_g\backslash G$: for regular g, the measure dg/dt of Part I, §4.1; for unipotent g, a fixed G-invariant measure on the class of g, used both in the Shalika expansion (Corollary 3.7) and in the Richardson formula of Lemma 5.10. "Same orbital integrals" below always means equality for these measures.

Fix $M \ge M_1$, and let $C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen},M}$ be the space of $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ supported in $p^{-1}(ZV_M)$, and $C_c^\infty(G/Z)_M$ the space of $\varphi \in C_c^\infty(G/Z)$ supported in ZV_M.

Proposition 3.6 (Transport)**.** ***[P]***

(a) Every element of $p^{-1}(ZV_M)$ is of the form $z\,\sigma(g)$ with $z \in \overline{Z}$ and $g \in V_M$. Every regular element of $p^{-1}(ZV_M)$ is good.

(b) The formula $(\mathcal{T}\tilde{f})(p(z)g) := \tilde{\theta}(z)\,\tilde{f}(z\,\sigma(g))$ ($z \in \overline{Z}$, $g \in V_M$) defines a linear bijection

$$\mathcal{T} : C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen},M} \xrightarrow{\sim} C_c^\infty(G/Z)_M,$$

with inverse $(\mathcal{T}^{-1}\varphi)(z\sigma(g)) = \tilde{\theta}(z)^{-1}\varphi(p(z)g)$ on $p^{-1}(ZV_M)$ and 0 elsewhere.

(c) For $z \in \overline{Z}$ and regular $\gamma \in V_M$,

$$\Phi(z\,\sigma(\gamma), \tilde{f}) = \tilde{\theta}(z)^{-1}\,\Phi(p(z)\gamma, \mathcal{T}\tilde{f}), \qquad F(z\,\sigma(\gamma), \tilde{f}) = \tilde{\theta}(z)^{-1}\,F(p(z)\gamma, \mathcal{T}\tilde{f}),$$

where on the right Φ and $F = \Delta\Phi$ are the linear orbital integrals with the same measures. More generally, the first identity holds for every $\gamma \in V_M$, with Φ the integral over $G_\gamma\backslash G$ (for instance, for unipotent γ).

(d) For a function β on G that is constant on conjugacy classes and on Z-cosets, with $\beta \cdot 1_{ZV_M}$ locally constant, $\mathcal{T}\big((\beta \circ p)\tilde{f}\big) = \beta \cdot \mathcal{T}\tilde{f}$.

Proof. (a) If $\tilde{x} \in p^{-1}(ZV_M)$, write $p(\tilde{x}) = ag$ with $g \in V_M$, and choose a lift z of a; then $\tilde{x} = \zeta z\sigma(g)$ for some $\zeta \in \mu_n \subset \overline{Z}$. A regular element $z\sigma(\gamma)$ is good by Proposition 3.3(b), since z is central.

(b) *Independence of the lift z of $a = p(z)$:* replacing z by ζz multiplies $\tilde{\theta}(z)$ by ζ and $\tilde{f}(z\sigma(g))$ by ζ^{-1}. *Independence of the factorization:* if $ag = a'g'$ with $g, g' \in V_M$, then $g' =$

cg with $c = a/a' \in 1 + \mathfrak{p}^{\lceil M/r \rceil}$ (Lemma 3.5(a)), so $\sigma(g') = \sigma(c)\sigma(g)$ (Proposition 3.3(c)). With z' a lift of a', $z := z'\sigma(c)$ is a lift of a, and

$$\tilde{\theta}(z')\tilde{f}(z'\sigma(g')) = \tilde{\theta}(z')\tilde{f}(z\sigma(g)) = \tilde{\theta}(\sigma(c))^{-1}\tilde{\theta}(z)\tilde{f}(z\sigma(g)) = \tilde{\theta}(z)\tilde{f}(z\sigma(g)),$$

by Lemma 3.5(b). So $\mathcal{T}\tilde{f}$ is a well-defined function on ZV_M, extended by 0. It is Z-invariant by construction, locally constant because σ is continuous and $\tilde{f}$ is locally constant, and supported in the set ZV_M, which is open and closed (Lemma 3.1(a)); its support is compact modulo Z because that of $\tilde{f}$ is compact modulo $\overline{Z}$. The inverse is well defined by the same computation, and it is smooth because $p^{-1}(ZV_M)$ is open and closed.

(c) For $x \in G$, $\tilde{x}^{-1}z\sigma(\gamma)\tilde{x} = z\,\sigma(x^{-1}\gamma x)$ by Proposition 3.3(a). Hence $\tilde{f}(\tilde{x}^{-1}z\sigma(\gamma)\tilde{x}) = \tilde{\theta}(z)^{-1}(\mathcal{T}\tilde{f})(x^{-1}p(z)\gamma x)$. Integrate over $T_\gamma\backslash G$, and note that $\Delta(p(z)\gamma) = \Delta(\gamma)$. For non-regular $\gamma \in V_M$ the same computation applies over $G_\gamma\backslash G$, since $\sigma(\gamma)$ commutes with $p^{-1}(G_\gamma)$ (Proposition 3.3(b)).

(d) is immediate from the definitions. □

3.4 The germ expansion at central good elements

We record what the transport says about germs. This is not needed for the matching theorem, but it identifies the germs of [FK, §8] and [KP2, §1] at central elements.

Corollary 3.7 (Germ expansion at central good elements)**.** ***[P]*** *(using* ***[E]****) Let* $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ *with* $G = \mathrm{GL}(r, F)$*, and let* $z \in \overline{Z}$*. There is a neighborhood* V' *of* 1 *such that for regular* $\gamma \in V'$*,*

$$\Phi(z\,\sigma(\gamma), \tilde{f}) = \sum_{\mathcal{O}} \Gamma_{\mathcal{O}}(\gamma)\,\Phi(z\,\sigma(u_{\mathcal{O}}), \tilde{f}),$$

where $\mathcal{O}$ *runs over the unipotent conjugacy classes of* $\mathrm{GL}(r, F)$ *(partitions of* r*),* $u_{\mathcal{O}} \in \mathcal{O}$*, and the* $\Gamma_{\mathcal{O}}$ *are the Shalika germs of* $\mathrm{GL}(r, F)$ *for the measures fixed above. On the D side the only unipotent class is* $\{1\}$*, and the expansion is Part I, Proposition 4.6.*

Proof. Apply Proposition 3.6 to $\tilde{f} \cdot 1_{p^{-1}(ZV_M)}$, and the Shalika germ expansion [Sha72], [HC70] **[E]** to $\mathcal{T}(\tilde{f} \cdot 1_{p^{-1}(ZV_M)})$ at 1. Since that function is Z-invariant, its expansion at $p(z)$ is its expansion at 1. By Proposition 3.6(c), applied also to the unipotent orbits, the factors $\tilde{\theta}(z)^{\pm 1}$ cancel. □

Remark 3.8. (1) [FK, §8, Theorem] and [KP2, Theorem 1.4] (credited there to Shalika, Harish-Chandra, Vignéras and Kazhdan–Patterson) give a germ expansion of genuine orbital integrals at every semisimple element, with germs indexed by the good elements above it, and a converse characterizing orbital-integral functions [KP2, Theorem 1.7]. The germs are shown to exist; their structure is not determined there ([KP2, §1, Remark 2]). Corollary 3.7 identifies them, at central good elements, with the Shalika germs of $\mathrm{GL}(r, F)$. In comparing normalizations, note that [FK] normalize $F = \Delta\Phi$ with Δ formed from distinct eigenvalues, so that at $z\,\sigma(u)$ with z central $F = |\det z|^{-(r-1)/2}\Phi$.

(2) The principle that orbital integrals on covers near a semisimple element reduce to the Lie algebra of its centralizer is part of W.-W. Li's semisimple descent [Li2, §4], and we make no claim of priority for Corollary 3.7. What we use is the explicit transport of Proposition 3.6, which is compatible with *both* covers and with $\jmath$ (Proposition 3.3(f)).

4 Orbital integrals of arbitrary genuine functions

This section extends the facts about orbital integrals on elliptic tori in Part I, Appendix A, from regular-supported functions to arbitrary genuine functions. For r prime, the only singular elements near which the elliptic regular set accumulates are central (Lemma 4.1), so once the behavior near $\overline{Z}$ is removed, the elliptic orbital integrals of an arbitrary genuine function are admissible in the sense of Part I, §4.3 (Lemma 4.2). The section depends on Part I, Appendix A, and introduces no new notation.

Lemma 4.1 (Singular elements, r prime)**.** ***[P]****Let $G = \mathrm{GL}(r, F)$ with r prime. A non-central singular semisimple element s has a neighborhood containing no regular elliptic element. Hence, for every elliptic maximal torus T, every point of $T \smallsetminus Z$ is regular, and $(T \smallsetminus ZV_M)/Z$ is a compact subset of T^{reg}/Z.*

Proof. Write $P_s = \prod_i Q_i^{m_i}$ with distinct monic irreducible Q_i. If there were only one factor, $P_s = Q^m$ with $m \deg Q = r$, so $m = 1$ (then P_s is separable and s is regular) or $m = r$ and $\deg Q = 1$ (then the semisimple element s has a single eigenvalue in F and is central). Both are excluded, so $P_s = AB$ with A, B coprime and of positive degree. By Hensel's lemma, every monic polynomial close enough to P_s factors as $A'B'$ with $\deg A' = \deg A$ and $\deg B' = \deg B$. So elements close to s have reducible characteristic polynomials and are not elliptic. The second statement holds because the singular elements of T are central, T/Z is compact, and $T \smallsetminus ZV_M$ is closed. □

Lemma 4.2 (Extension of Part I, Lemma A.3)**.** ***[P]*** *(using* ***[E]****) Let $\tilde{f} \in C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen}}$ be arbitrary, T an elliptic maximal torus, and $h = F(\cdot, \tilde{f})$ on $\overline{T}^{g,\mathrm{reg}}$. Then h satisfies conditions (i), (ii) and (iv) of Part I, §4.3. If moreover $h = 0$ on $p^{-1}(ZV_M) \cap \overline{T}^{g,\mathrm{reg}}$ for some M, then h is admissible.*

Proof. The proofs of (ii) and (iv) in Part I, Lemma A.3, do not use the support of $\tilde{f}$. The proof of (i) uses Part I, Lemma A.1(b), with $\Omega = p(\mathrm{supp}\, \tilde{f})$; that lemma holds for every Ω compact modulo Z [HC70] **[E]**. For (iii): $\mathrm{supp}\, h$ is closed in the open and closed subgroup $Z(\overline{T})$ and contained in $p^{-1}(T \smallsetminus ZV_M)$, whose image in T/Z is compact and regular by Lemma 4.1. □

Remark 4.3. Lemma 4.1 is the only place where primality of r enters the local theory of this paper (besides its use in Part I). For composite r there are non-central singular semisimple elements with elliptic centralizers, of the form $\mathrm{GL}(m, E')$ with $m[E' : F] = r$ and $m > 1$. Germ expansions there exist by [FK, §8], but they are not explicit, and the method of §3 would have to be extended to covers of $\mathrm{GL}(m, E')$.

5 Matching of arbitrary functions

This section proves Theorem E. We first collect the inputs about the linear groups (§5.1). We then construct the genuine analog of the pair (constant function on $D_v^\times$, Steinberg pseudo-coefficient on $\mathrm{GL}(r, F)$) near the center (§5.2), prove matching in both directions

(§5.3), and determine the germ near the center that every matching function must have (§5.4). Throughout, $v \in S$, $F = K_v$, and $\tilde{\theta}_v$ is a genuine character of $\overline{Z}_v$, common to both covers. The section depends on §§3–4, on Part I, §4, and on the linear results of [FK], [Kaz] and [Bad] cited in §5.1. It introduces the linear inputs (L1), (L2), the functions f_{St}, e_D, $\tilde{f}^{\mathrm{St}}$, and the conditions (Ell), (Non), (G1), (G2).

5.1 Inputs about the linear groups

We state the two linear inputs (L1) and (L2), which are used throughout this section, prove the value of the Steinberg pseudo-coefficient at the identity, and derive (L2) from statements in [FK] and [Bad].

Let $G = \mathrm{GL}(r, F)$.

(L1) **[E]** There is $f_{\mathrm{St}} \in C_c^\infty(G/Z)$, a pseudo-coefficient of the Steinberg representation, with $\operatorname{tr}\mathrm{St}(f_{\mathrm{St}}) = 1$, $\operatorname{tr}\pi'(f_{\mathrm{St}}) = 0$ for every irreducible tempered $\pi' \not\cong \mathrm{St}$ with trivial central character, $\Phi(\gamma, f_{\mathrm{St}}) = (-1)^{r-1}\operatorname{vol}(T_\gamma/Z)^{-1}$ for regular elliptic γ, and $\Phi(\gamma, f_{\mathrm{St}}) = 0$ for regular non-elliptic γ.

(L2) **[P]** (Lemma 5.3) Every $\varphi \in C_c^\infty(G/Z)$ with $\Phi(\gamma, \varphi) = 0$ for all regular non-elliptic γ has a matching $\varphi_D \in C_c^\infty(D_v^\times/Z)$.

Source of (L1). [FK, §25.1, Theorem], from [Kaz, Theorem K], is stated for every square-integrable representation of the cover, and so, with $n = 1$ (the convention of [FK, §8, Remark (3)]), for $\mathrm{GL}(r, F)$. It gives $\Phi^z(\gamma, f_{\mathrm{St}}) = \overline{\Theta_{\mathrm{St}}(\gamma)}$ on the regular elliptic set and $\Phi^z = 0$ at regular non-elliptic elements, together with the trace properties. Since $\Phi^z = \operatorname{vol}(T_\gamma/Z)\Phi$ and $\Theta_{\mathrm{St}} = (-1)^{r-1}$ on the regular elliptic set, this is (L1). The value of Θ_{St} follows from [Bad, Theorem 2.2] with $n = 1$, $d = r$, since the trivial representation of $D_v^\times$ corresponds to St.

Remark 5.1. **[P]** *From compact support to trivial central character.* The linear statements (L1), (L2) are usually formulated for $C_c^\infty(G)$ with measures on G and T. Choose $b \in C_c^\infty(F^\times)$ supported in small neighborhoods of representatives of $F^\times/F^{\times r}$ (a finite group), normalized so that $\int_{F^\times} b(z^r y)\, dz = 1$ for all y. For $\varphi \in C_c^\infty(G/Z)$, the function $\varphi' := \varphi \cdot (b \circ \det)$ (respectively $b \circ \mathrm{Nrd}$) lies in $C_c^\infty(G)$, satisfies $\int_Z \varphi'(zg)\, dz = \varphi(g)$, and has orbital integrals $b(\det\gamma)$ times those of φ. In particular φ' is cuspidal if φ is. Since averaging over Z commutes with matching ($z\gamma \leftrightarrow z\gamma'$, $\Delta(z\gamma) = \Delta(\gamma)$), matching statements for C_c^∞ imply the same statements for $C_c^\infty(\cdot/Z)$.

Lemma 5.2 (Value of f_{St} at the identity)**.** ***[P]*** *(using **[E]**) With f_{St} as in (L1), $f_{\mathrm{St}}(1) = d(\mathrm{St})$, the formal degree of* St *with respect to dg.*

Proof. By Harish-Chandra's Plancherel formula for G/Z with trivial central character [Wal03, Théorème VIII.1.1, in the form (3) of §VIII.1] (in explicit form for $\mathrm{GL}(n)$, see [AP]), for $f \in C_c^\infty(G/Z)$,

$$f(1) = \sum_\pi d(\pi)\operatorname{tr}\pi(f) + \sum_{P \neq G} \int \operatorname{tr}\left(\mathrm{Ind}_P^G \sigma\right)(f)\, d\mu_P(\sigma),$$

where π runs over square-integrable representations with trivial central character, and the second sum runs over proper standard parabolic subgroups $P = MN$, with σ running over unitary twists of square-integrable representations of M and μ_P absolutely continuous Plancherel measures. For $\mathrm{GL}(r, F)$ every $\mathrm{Ind}_P^G \sigma$ here is irreducible and tempered (the segments attached to unitary twists of square-integrable representations are pairwise unlinked, so [Zel80, Theorem 9.7(a)] applies), and it is not square-integrable when $P \neq G$, so it is not St. By [FK, §25.1, Theorem] with $n = 1$, $\operatorname{tr} \pi'(f_{\mathrm{St}}) = 0$ for every irreducible tempered $\pi' \ncong \mathrm{St}$, and $\operatorname{tr} \mathrm{St}(f_{\mathrm{St}}) = 1$. Hence only the term $\pi = \mathrm{St}$ survives, and $f_{\mathrm{St}}(1) = d(\mathrm{St})$. □

Lemma 5.3 (The linear input (L2))**.** ***[P]*** *(using **[E]**) Let $G = \mathrm{GL}(r, F)$, r prime, and let $\varphi \in C_c^\infty(G/Z)$ satisfy $\Phi(\gamma, \varphi) = 0$ for all regular non-elliptic γ. Then there is $\varphi_D \in C_c^\infty(D_v^\times/Z)$ with $F(t, \varphi_D) = (-1)^{r-1} F(j(t), \varphi)$ for every regular $t \in D_v^\times$ and every embedding j of $F[t]$ in $M_r(F)$.*

Proof. Step 1 (finitely many discrete series). Let φ be bi-invariant under an open compact subgroup K'. Then $\operatorname{tr} \pi(\varphi) \neq 0$ only if $\pi^{K'} \neq 0$. There are only finitely many square-integrable π with trivial central character and $\pi^{K'} \neq 0$: by Harish-Chandra's finiteness theorem [Wal03, Théorème VIII.1.2] **[E]** these form finitely many orbits under twisting by unramified characters $|\det|^s$, and in each orbit only finitely many members have trivial central character (the twist multiplies the central character by $|z|^{rs}$). Let Π be this finite set, and for $\pi \in \Pi$ let f_π be a pseudo-coefficient as in [FK, §25.1, Theorem] with $n = 1$ **[E]**. Put

$$\psi := \varphi - \sum_{\pi \in \Pi} \operatorname{tr} \pi(\varphi)\, f_\pi.$$

Step 2 (ψ has vanishing tempered traces). Let π' be irreducible and tempered, with trivial central character. If π' is square-integrable, then $\operatorname{tr} \pi'(\psi) = \operatorname{tr} \pi'(\varphi) - \operatorname{tr} \pi'(\varphi) = 0$ by the defining property of the f_π (and $\operatorname{tr} \pi'(\varphi) = 0$ if $\pi' \notin \Pi$). Otherwise $\pi' = \mathrm{Ind}_P^G \sigma$ with $P \neq G$ and σ a unitary twist of a square-integrable representation of M (the classification of tempered representations of $\mathrm{GL}(r, F)$; [Zel80, §9]), and $\operatorname{tr} \pi'(f_\pi) = 0$ by [FK, §25.1]. Also $\operatorname{tr} \pi'(\varphi) = 0$: by the Weyl integration formula (characters are locally integrable [FK, §13]), $\operatorname{tr} \pi'(\varphi)$ is a sum over conjugacy classes of maximal tori of integrals of $\Theta_{\pi'} \cdot \Phi(\cdot, \varphi)$. On non-elliptic tori $\Phi(\cdot, \varphi) = 0$ by hypothesis, and on elliptic tori $\Theta_{\pi'} = 0$, since the character of a representation induced from a proper parabolic subgroup vanishes on the regular elliptic set [vD72] **[E]**.

Step 3 (density). We use [FK, §27.3, Proposition] **[E]** with $n = 1$. For $n = 1$ the cover is G itself, $x^* = x$, every representation is genuine and relevant, and the correspondence of [FK, §26] is the identity (the character relation (26.1) becomes $\Theta_{\tilde{\pi}} = \Theta_\pi$); "matching" means equality of $F(x, \cdot)$ at regular x. The proposition then reads: *if $f, f' \in C_c^\infty(G/Z)$ satisfy $\operatorname{tr} \pi(f) = \operatorname{tr} \pi(f')$ for every irreducible tempered π with trivial central character, then $F(x, f) = F(x, f')$ for every regular x.* Apply it with $f = \psi$ and $f' = 0$, using Step 2. (For covers in general, the corresponding density of tempered characters is proved in [Li2].) So $\Phi(\gamma, \varphi) = \sum_{\pi \in \Pi} \operatorname{tr} \pi(\varphi) \Phi(\gamma, f_\pi)$ for all regular γ, and on the regular elliptic set

$$\Phi^z(\gamma, \varphi) = \sum_{\pi \in \Pi} c_\pi \overline{\Theta_\pi(\gamma)}, \qquad c_\pi := \operatorname{tr} \pi(\varphi).$$

Step 4 (transfer). By [Bad, Theorem 2.2] with $n = 1$, $d = r$ **[E]**, there is a bijection $\pi \mapsto C(\pi)$ from essentially square-integrable representations of $\mathrm{GL}(r, F)$ to irreducible representations of $D_v^\times$, preserving central characters, with $\Theta_{C(\pi)}(t) = (-1)^{r-1}\Theta_\pi(j(t))$ for regular t. For an irreducible representation τ of $D_v^\times$ with trivial central character, put $f_\tau := \mathrm{vol}(D_v^\times/Z)^{-1}\,\overline{\Theta_\tau} \in C_c^\infty(D_v^\times/Z)$. Since f_τ is conjugation invariant, $\Phi^z(t, f_\tau) = \mathrm{vol}(D_v^\times/Z) f_\tau(t) = \overline{\Theta_\tau(t)}$. Put $\varphi_D := \sum_{\pi\in\Pi} c_\pi f_{C(\pi)}$. Then

$$\Phi^z(t, \varphi_D) = \sum_\pi c_\pi \overline{\Theta_{C(\pi)}(t)} = (-1)^{r-1} \sum_\pi c_\pi \overline{\Theta_\pi(j(t))} = (-1)^{r-1}\Phi^z(j(t), \varphi).$$

Divide by $\mathrm{vol}(T/Z)$, which is the same on both sides because the same dt is used on T and $j(T)$ (Part I, §4.1), and multiply by $\Delta(t) = \Delta(j(t))$. The measure $dg_{D,v}$ enters only through $\mathrm{vol}(D_v^\times/Z)$ in f_τ and in $\Phi^z(t, \cdot)$, and cancels.

On the complex conjugate. The identity $\Phi^z(\gamma, f_\pi) = \overline{\Theta_\pi(\gamma)}$ is the normalization compatible with $\mathrm{tr}\,\pi(f_\pi) = 1$: by the Weyl integration formula, for f with vanishing non-elliptic orbital integrals, $\mathrm{tr}\,\pi'(f) = \sum_T (|W_T|\,\mathrm{vol}(T/Z))^{-1} \int_{T/Z} \Delta(t)^2 \Theta_{\pi'}(t)\Phi^z(t, f)\,dt$, which for $\Phi^z(\cdot, f) = \overline{\Theta_\pi}$ is the elliptic inner product $\langle \Theta_{\pi'}, \Theta_\pi\rangle$ of [FK, §24], equal to $\delta_{\pi\pi'}$ for square-integrable π' [FK, §25]. In any case the argument is insensitive to this convention: replacing $\overline{\Theta}$ by Θ throughout Steps 3 and 4 changes nothing, because the transfer identity of Step 4 has real coefficients $(-1)^{r-1}$. □

Remark 5.4. Lemma 5.3, combined with Remark 5.1, is the converse half of the linear transfer of functions of [DKV]. We include the proof because [Bad, §4.6] states only the regular-supported case, and because it uses only results whose statements we have checked in [FK] and [Bad].

5.2 The Steinberg-type function

We construct the genuine analog, near the center, of the pair (constant function on $D_v^\times$, Steinberg pseudo-coefficient). It exists as a pair of functions with matching orbital integrals; it is not a pair of pseudo-coefficients of single genuine representations. The functions e_D and $\tilde{f}^{\mathrm{St}}$ defined here enter Theorem E.

Fix $M \geq M_1$ (for $\tilde{\theta}_v$, on both sides) and put

$$e_D := \mathcal{T}_D^{-1}\big(1_{ZV_{D,M}}\big) \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}},$$
$$\tilde{f}^{\mathrm{St}} := d(\mathrm{St})^{-1}\,\mathcal{T}^{-1}\big(f_{\mathrm{St}} \cdot 1_{ZV_M}\big) \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}},$$

with f_{St} as in (L1). Here $1_{ZV_{D,M}} \in C_c^\infty(D_v^\times/Z)$ because $D_v^\times/Z$ is compact and $ZV_{D,M}$ is open and closed. Concretely, $e_D(z\sigma_D(g)) = \tilde{\theta}_v(z)^{-1}$ for $g \in V_{D,M}$, and $e_D = 0$ off $p^{-1}(ZV_{D,M})$. So $e_D(1) = 1$.

Proposition 5.5. ***[P]*** *(using (L1)) The functions e_D and $\tilde{f}^{\mathrm{St}}$ match. Explicitly, $\tilde{f}^{\mathrm{St}}(1) = 1 = e_D(1)$, and at good regular elements*

$$F(z\sigma(\gamma), \tilde{f}^{\mathrm{St}}) = \begin{cases} (-1)^{r-1}\tilde{\theta}_v(z)^{-1}\Delta(\gamma)\big(d(\mathrm{St})\,\mathrm{vol}(T_\gamma/Z)\big)^{-1} & \gamma \in V_M \text{ regular elliptic}, \\ 0 & \gamma \in V_M \text{ regular non-elliptic}, \end{cases}$$

and $F(\cdot, \tilde{f}^{\mathrm{St}}) = 0$ *at good regular elements outside* $p^{-1}(ZV_M)$.

Proof. By Proposition 3.6(c),(d), $F(z\sigma(\gamma), \tilde{f}^{\mathrm{St}}) = d(\mathrm{St})^{-1}\tilde{\theta}_v(z)^{-1}1_{V_M}(\gamma)\Delta(\gamma)\Phi(\gamma, f_{\mathrm{St}})$, and (L1) gives the displayed values. On the D side, $e_D(\tilde{x}^{-1}z\sigma_D(t)\tilde{x}) = \tilde{\theta}_v(z)^{-1}1_{V_{D,M}}(t)$, so $F(z\sigma_D(t), e_D) = \tilde{\theta}_v(z)^{-1}1_{V_{D,M}}(t)\Delta(t)\,\mathrm{vol}(D_v^\times/Z)/\,\mathrm{vol}(T/Z)$, and $\mathrm{vol}(D_v^\times/Z) = d(\mathrm{St})^{-1}$ by Definition 2.1. Outside $p^{-1}(ZV_{D,M})$ and $p^{-1}(ZV_M)$ both orbital integrals vanish, and $\jmath$ preserves these sets, since it preserves characteristic polynomials and is the identity on $\overline{Z}_v$. Now compare, using Proposition 3.3(f), the equality $1_{V_{D,M}}(t) = 1_{V_M}(j(t))$, and the sign $(-1)^{r-1}$ in (M1). □

Remark 5.6 (Not a pseudo-coefficient of one genuine representation)**.** $\tilde{f}^{\mathrm{St}}$ is defined by orbital integrals, not by representations, and matching needs nothing more. Under the character relations (D2) of Part I (proved at tame places where the corrected Flicker–Kazhdan correspondence is available, for example when $(n, N_r) = 1$; see §7(ii)), $\mathrm{tr}\,\overline{\pi}(\tilde{f}^{\mathrm{St}}) = \mathrm{tr}\,\overline{\pi}_D(e_D)$ for corresponding pairs. Now $\mathrm{tr}\,\tau(e_D) = \int_{\overline{G}_D(K_v)/\overline{Z}_v} e_D\,\Theta_\tau$, which is the integral of $g \mapsto \Theta_\tau(\sigma_D(g))$ over $ZV_{D,M}/Z$, is in general nonzero for more than one genuine irreducible τ (Fourier inversion on the compact group $\overline{G}_D(K_v)/\overline{Z}_v$ expresses e_D in terms of the characters of all τ of level contained in $V_{D,M}$). So $\tilde{f}^{\mathrm{St}}$ should be a combination of pseudo-coefficients of several genuine square-integrable representations, not the pseudo-coefficient of a single "genuine Steinberg" representation. Accordingly, the phrase "a genuine analog of a Steinberg pseudo-coefficient" in Part I, §4.5, should be read as "a genuine function with the orbital integrals of a Steinberg pseudo-coefficient near the center".

Remark 5.7 (No genuine characters when $v \mid n$)**.** **[P]**If $v \mid n$, then $\overline{G}_D(K_v)$ has no genuine one-dimensional representation. By Part I, Proposition 3.5(b), some covered elliptic torus $\overline{E}_D$ is non-abelian, so there are commuting $x, y \in E^\times$ whose lifts have commutator $\zeta \neq 1$. A one-dimensional genuine χ would give $1 = \chi([\tilde{x}, \tilde{y}]) = \zeta$. So the linear pair (trivial representation of $D^\times$, St) has no direct analog among genuine representations. The pair $(e_D, \tilde{f}^{\mathrm{St}})$ is its analog among functions.

5.3 The matching theorem

We now prove Theorem E. The D-side function is split into a multiple of e_D and a function vanishing near the center; the GL-side function is split by a cut-off into a piece near the center and a piece away from it.

Theorem 5.8 (Matching of arbitrary functions)**.** ***[P]*** *(using (L1), (L2)) Let* $v \in S$.

(a) Every $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ *has a matching* $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$. *One may take*

$$\tilde{f} = \tilde{f}_D(1)\,\tilde{f}^{\mathrm{St}} + \tilde{f}_{\mathrm{reg}},$$

with $\tilde{f}_{\mathrm{reg}}$ *supported on the good regular elliptic set. This uses only (L1).*

(b) Every $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ *with* $F(\tilde{\delta}, \tilde{f}) = 0$ *for all good regular non-elliptic* $\tilde{\delta}$ *has a matching* $\tilde{f}_D$. *This uses (L2).*

Proof. (a) By construction $e_D(z) = \tilde{\theta}_v(z)^{-1}$ on $\overline{Z}_v$, so $\tilde{f}_D - \tilde{f}_D(1)e_D$ vanishes on $\overline{Z}_v$. Being smooth with support compact modulo $\overline{Z}_v$, it is right invariant under some $\kappa(U)$, so it vanishes on the neighborhood $\overline{Z}_v\kappa(U)$ of $\overline{Z}_v$. By Part I, Theorem 4.5(a), it has a matching $\tilde{f}_{\mathrm{reg}}$ supported on the good regular elliptic set. By Proposition 5.5, e_D matches $\tilde{f}^{\mathrm{St}}$. Matching is linear.

(b) Fix $M \geq M_1$ and write $\tilde{f} = \tilde{f}_1 + \tilde{f}_2$ with $\tilde{f}_1 = \tilde{f} \cdot 1_{p^{-1}(ZV_M)}$. Both pieces are smooth (the set is open and closed), and both have vanishing non-elliptic orbital integrals, since the cut-off is constant on conjugacy classes (Proposition 3.6(d)).

The piece near the center. $\varphi := \mathcal{T}\tilde{f}_1 \in C_c^\infty(G/Z)_M$ has vanishing regular non-elliptic orbital integrals, by Proposition 3.6(c) and because orbital integrals of φ outside ZV_M vanish. By (L2) and Remark 5.1, there is a matching $\varphi_D \in C_c^\infty(D_v^\times/Z)$. Since the elliptic orbital integrals of φ vanish off ZV_M, $\varphi_D \cdot 1_{ZV_{D,M}}$ still matches φ, and it lies in $C_c^\infty(D_v^\times/Z)_M$. By Propositions 3.6(c) and 3.3(f), $\tilde{f}_{D,1} := \mathcal{T}_D^{-1}(\varphi_D 1_{ZV_{D,M}})$ matches $\tilde{f}_1$. (The good regular elements of $p^{-1}(ZV_{D,M})$ correspond under $\jmath$ to the good regular elliptic elements of $p^{-1}(ZV_M)$, as in the proof of Theorem 5.13(a).)

The piece away from the center. By Lemma 4.2, for each i the function $F(\cdot, \tilde{f}_2)$ on $Z(\overline{E}_{i,\mathrm{GL}})^{\mathrm{reg}}$ is admissible, since it vanishes on $p^{-1}(ZV_M)$. Transport by $\jmath_i^{-1}$, multiply by $(-1)^{r-1}$, realize on $\overline{G}_D(K_v)$ by Part I, Lemma 4.4(b),(c), and sum over i, exactly as in the proof of Part I, Theorem 4.5(b). This gives $\tilde{f}_{D,2}$, and (M2) for $\tilde{f}_2$ is the hypothesis.

Then $\tilde{f}_D := \tilde{f}_{D,1} + \tilde{f}_{D,2}$ matches $\tilde{f}$. □

Remark 5.9. (1) Theorem 5.8(b) applies to supercusp forms at $v \in S$, whose regular non-elliptic orbital integrals vanish (Selberg's principle; [FK, §13(i)]) **[E]**. This is item (D1)(b) of Part I.

(2) An alternative proof of (b), using (L1) and Lemma 5.10 instead of (L2), is noted after Theorem 5.13.

(3) Theorem 5.8 does not by itself give $\operatorname{tr}\overline{\pi}(\tilde{f}) = \operatorname{tr}\overline{\pi}_D(\tilde{f}_D)$ for single representations; that is goal (ii) (§7). It does strengthen the global results of Part I, since arbitrary test functions may now be used at the places of $S \smallsetminus \{u\}$ (§6).

5.4 The germ near the center

We determine the orbital integrals near $\overline{Z}_v$ that any function matching a given $\tilde{f}_D$ must have. The point is that $\tilde{f}_D$ has only one number's worth of germ at the whole center, namely $\tilde{f}_D(1)$. The conditions (Ell), (Non), (G1), (G2) and the germs $\Gamma_{\mathcal{O}}$ appear here.

Let $v \in S$ and $\tilde{f}_D \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$. Since $\tilde{f}_D(z\tilde{g}) = \tilde{\theta}_v(z)^{-1}\tilde{f}_D(\tilde{g})$, we have $\tilde{f}_D(z) = \tilde{\theta}_v(z)^{-1}\tilde{f}_D(1)$ on $\overline{Z}_v$. So Part I, Proposition 4.6, together with Proposition 3.3(d), reads

$$F(z\,\sigma_D(t), \tilde{f}_D) = \tilde{\theta}_v(z)^{-1}\,\tilde{f}_D(1)\,\frac{\Delta(t)}{d(\mathrm{St})\,\operatorname{vol}(T/Z)} \qquad (t \in T \cap V_{D,M} \text{ regular}), \tag{1}$$

for M large (depending on $\tilde{f}_D$), every maximal subfield E, $T = E^\times$, and $z \in \overline{Z}_v$. We used $\operatorname{vol}(T\backslash D_v^\times) = \operatorname{vol}(D_v^\times/Z)/\operatorname{vol}(T/Z)$ and Definition 2.1. Note that (1) is uniform over all types of maximal subfields, cyclic or not.

We first prove two facts about the linear group. Let $G = \mathrm{GL}(r, F)$ (here $r \geq 2$ need not be prime). Shalika germs $\Gamma_{\mathcal{O}}$ are normalized so that $\Phi(\gamma, f) = \sum_{\mathcal{O}} \Gamma_{\mathcal{O}}(\gamma)\Phi(u_{\mathcal{O}}, f)$ for regular γ near 1, with $\Phi(1, f) = f(1)$, unnormalized orbital integrals Φ, and fixed invariant measures on the unipotent classes [Sha72].

Lemma 5.10 (Functions cuspidal near the identity)**.** ***[P]*** *(using* ***[E]****) Let $\varphi \in C_c^\infty(G/Z)$, and let Ω be a conjugation-invariant neighborhood of 1 in G. If $\Phi(\gamma, \varphi) = 0$ for every regular non-elliptic $\gamma \in \Omega$, then $\Phi(u, \varphi) = 0$ for every non-trivial unipotent $u \in G$. Consequently $\Phi(\gamma, \varphi) = \Gamma_{\{1\}}(\gamma)\,\varphi(1)$ for regular γ near 1.*

Proof. Step 1 (compact support). Choose $\eta \in C_c^\infty(F^\times)$ equal to 1 near 1, and put $f(g) := \varphi(g)\eta(\det g)$. Then $f \in C_c^\infty(G)$, since $z \mapsto z^r$ is proper on $F^\times$. The unipotent orbital integrals of f and φ agree, because $\det u = 1$ on unipotent classes; and the regular orbital integrals agree near 1, because $\eta \circ \det = 1$ on the conjugacy class of γ when $\det \gamma$ is near 1. So we may replace φ by f and Ω by a smaller neighborhood.

Step 2 (descent). Let $P = MN$ be a proper standard parabolic subgroup, and $f^P(m) = \delta_P(m)^{1/2} \int_K \int_N f(k^{-1}mnk)\,dn\,dk$ the constant term, with $K = \mathrm{GL}(r, \mathcal{O})$. By [FK, §7, Proposition] with $n = 1$ [**E**], $F(x, f) = F^M(x, f^P)$ for every $x \in M$ that is regular in G. Here $F = \Delta\Phi$ and $F^M = \Delta_M\Phi^M$ are normalized orbital integrals, in the conventions of §2.1, and [FK] define the constant term with the same factor $\delta_P^{1/2}$ as here. Since $\Delta(x)$ and $\Delta_M(x)$ are nonzero at such x, $\Phi(x, f) = 0$ if and only if $\Phi^M(x, f^P) = 0$. Compare the parabolic descent statement [KT, Proposition 2.39]; its differently normalized descent function differs from ours by factors that do not vanish (discriminants, and $\delta_P(x) = 1$ for x near 1), so it gives the same vanishing assertion. Such x are non-elliptic in G, since their centralizer lies in M. So $\Phi^M(x, f^P) = 0$ for $x \in M \cap \Omega$ regular in G. The elements of M that are regular in G are dense in the M-regular set near 1 (scale the blocks by distinct central scalars close to 1), and $\Phi^M(\cdot, f^P)$ is locally constant on the M-regular set. Hence $\Phi^M(x, f^P) = 0$ for all M-regular x near 1.

Step 3 (independence of germs on M). By the Shalika expansion on $M \simeq \prod_i \mathrm{GL}(n_i, F)$ [Sha72] [**E**], $0 = \sum_\nu \Gamma_\nu^M(x)\Phi^M(u_\nu, f^P)$ for M-regular x near 1, the sum over unipotent classes ν of M. The germs Γ_ν^M are products of the germs of the factors, and the germs of each $\mathrm{GL}(n_i, F)$ are linearly independent as germs at 1 on the regular set ([KP2, Proposition 1.6] with $n = 1$; [HC78, Lemma 24]) [**E**]. Products of linearly independent families on the factors are linearly independent on the product. So every $\Phi^M(u_\nu, f^P)$ vanishes; in particular $f^P(1) = 0$.

Step 4 (Richardson classes). Let u be a non-trivial unipotent element with Jordan type $\lambda \neq (1^r)$, and let $P_\lambda = M_\lambda N_\lambda$ be the standard parabolic subgroup whose Levi blocks have the sizes of the transposed partition λ^t. It is proper, and the class of u is the Richardson class of P_λ, the dense orbit in N_λ. We claim that $\Phi(u, f) = a_\lambda f^{P_\lambda}(1)$ with $a_\lambda > 0$ depending only on the normalizations of measures. On the Lie algebra this is [How74, Proposition 5] [**E**]: for $\phi \in C_c^\infty(\mathfrak{gl}_r(F))$ and the nilpotent orbit $\mathcal{O}_\lambda$, the orbital integral is $\int_{\mathfrak{u}_\lambda} \int_K \phi(\mathrm{Ad}(k)Y)\,dk\,dY$, where $\mathfrak{u}_\lambda$ is the nilradical of the parabolic subalgebra stabilizing the flag defined by an element of $\mathcal{O}_\lambda$ (that is, of $\mathrm{Lie}\,P_\lambda$, up to conjugacy). The map $Y \mapsto \exp Y = \sum_{k<r} Y^k/k!$ is a G-equivariant homeomorphism from the nilpotent cone onto the unipotent set, carrying $\mathcal{O}_\lambda$ onto the class of u and $\mathfrak{u}_\lambda$ onto

N_λ, with $\exp_* dY = dn$ on $\mathfrak{u}_\lambda$ (Haar measures on a unipotent group correspond under exp). Choose $\phi \in C_c^\infty(\mathfrak{gl}_r(F))$ with $\phi(Y) = f(\exp Y)$ for nilpotent Y (possible since the nilpotent cone is closed, exp is a homeomorphism onto the closed unipotent set, and so $f \circ \exp$ is locally constant with compact support on the cone; such functions extend to $C_c^\infty(\mathfrak{gl}_r(F))$). Howe's identity involves ϕ only at nilpotent elements. Then both sides of Howe's identity for ϕ are the corresponding integrals for f, which gives the claim. By Step 3, $\Phi(u, f) = 0$.

The last assertion follows from the Shalika expansion of φ at 1. □

Corollary 5.11 (The trivial germ)**.** ***[P]*** *(using (L1)) For regular γ in a neighborhood of 1,*

$$\Gamma_{\{1\}}(\gamma) = \begin{cases} (-1)^{r-1}\big(d(\mathrm{St})\,\mathrm{vol}(T_\gamma/Z)\big)^{-1} & \gamma \text{ elliptic}, \\ 0 & \gamma \text{ non-elliptic}. \end{cases}$$

Proof. By (L1), f_{St} satisfies the hypothesis of Lemma 5.10 with $\Omega = G$. Hence $\Phi(\gamma, f_{\mathrm{St}}) = \Gamma_{\{1\}}(\gamma) f_{\mathrm{St}}(1) = \Gamma_{\{1\}}(\gamma) d(\mathrm{St})$ near 1 (Lemma 5.2). Compare with the values of $\Phi(\gamma, f_{\mathrm{St}})$ in (L1). □

Remark 5.12. Corollary 5.11 agrees with Rogawski's theorem on the trivial germ on elliptic tori [Rog81] and with Harish-Chandra's vanishing of the trivial germ on non-elliptic tori [HC78]. Our proof derives it from (L1) and the Plancherel formula instead.

Theorem 5.13 (The required germ)**.** *Let $\tilde{f} \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$, and let $M \geq M_1$ be large enough for (1) on both sides.*

(a) ***[P]*** *$\tilde{f}$ satisfies (M1) and (M2) of Part I, Definition 4.3, with $\tilde{f}_D$ at all good regular elements of $p^{-1}(ZV_M)$ if and only if*

$$F(z\,\sigma(\gamma), \tilde{f}) = (-1)^{r-1}\,\tilde{\theta}_v(z)^{-1}\,\tilde{f}_D(1)\,\frac{\Delta(\gamma)}{d(\mathrm{St})\ \mathrm{vol}(T_\gamma/Z)} \quad (\gamma \in V_M \text{ reg. elliptic}), \tag{Ell}$$

$$F(z\,\sigma(\gamma), \tilde{f}) = 0 \quad (\gamma \in V_M \text{ reg. non-elliptic}), \tag{Non}$$

for all $z \in \overline{Z}_v$. The right side of (Ell) is the same multiple of $\Delta(\gamma)/\,\mathrm{vol}(T_\gamma/Z)$ for every elliptic torus.

(b) ***[P]*** *(Germ form.) There is $M' \geq M$, depending on $\tilde{f}$, such that (Ell) and (Non) hold for all regular $\gamma \in V_{M'}$ if and only if*

(G1) $\Phi(z\,\sigma(u), \tilde{f}) = 0$ for every non-trivial unipotent u and every $z \in \overline{Z}_v$, and

(G2) $\tilde{f}(1) = \tilde{f}_D(1)$.

(c) ***[P]*** *For arbitrary Haar measures at v, as allowed in Part I, (G2) would read $\tilde{f}(1) = \lambda_v \tilde{f}_D(1)$ with $\lambda_v = d(\mathrm{St})\,\mathrm{vol}(D_v^\times/Z)$ (Remark 2.2(3)).*

Proof. (a) Every good regular element of $p^{-1}(ZV_{D,M}) \subset \overline{G}_D(K_v)$ is conjugate to some $z\sigma_D(t)$ with $t \in T \cap V_{D,M}$ regular, for some maximal subfield E, $T = E^\times$ (Proposition 3.6(a)). By Proposition 3.3(f), $\jmath$ maps its class to that of $z\sigma_{\mathrm{GL}}(j(t))$, and $j(t) \in V_M$ because $P_{j(t)} = P_t$. Conversely, every good regular elliptic element of $p^{-1}(ZV_M) \subset \overline{G}_v$ arises in this way (Part I, Corollary 3.12). So (M1) at these elements is (Ell), by (1) and $\mathrm{vol}(T_{j(t)}/Z) = \mathrm{vol}(T/Z)$ (Part I, §4.1). (M2) at these elements is (Non).

(b) Put $\varphi = \mathcal{T}(\tilde{f}1_{p^{-1}(ZV_M)})$, so that $\varphi(1) = \tilde{f}(1)$. By Proposition 3.6(c), extended to non-regular elements, $\Phi(z\sigma(u), \tilde{f}) = \tilde{\theta}_v(z)^{-1}\Phi(p(z)u, \varphi) = \tilde{\theta}_v(z)^{-1}\Phi(u, \varphi)$ for unipotent u, since φ is Z-invariant. So (G1) says that $\Phi(u, \varphi) = 0$ for all non-trivial unipotent u.

Sufficiency. By Corollary 3.7 and (G1),

$$\Phi(z\sigma(\gamma), \tilde{f}) = \Gamma_{\{1\}}(\gamma)\tilde{f}(z) = \tilde{\theta}_v(z)^{-1}\Gamma_{\{1\}}(\gamma)\tilde{f}(1)$$

for regular γ near 1. Corollary 5.11 and (G2) give (Ell) and (Non) after multiplying by $\Delta(\gamma)$, near $\overline{Z}_v$.

Necessity. By (Non) and Proposition 3.6(c), $\Phi(\gamma, \varphi) = 0$ for regular non-elliptic $\gamma \in ZV_M$. Lemma 5.10 with $\Omega = ZV_M$ gives (G1). Then, as in the sufficiency part, $F(z\sigma(\gamma), \tilde{f}) = \tilde{\theta}_v(z)^{-1}\Delta(\gamma)\Gamma_{\{1\}}(\gamma)\tilde{f}(1)$ near $\overline{Z}_v$. Comparing with (Ell) at regular elliptic γ near 1, where $\Gamma_{\{1\}}(\gamma) \neq 0$ by Corollary 5.11, gives (G2).

(c) Repeat the computation in (b) with $\mathrm{vol}(D_v^\times/Z)$ in place of $d(\mathrm{St})^{-1}$ in (1) and (Ell). □

Remark 5.14. For matching, only part (a) of Theorem 5.13 is used. Part (b) is included because it states the result in terms of the germ expansion of Corollary 3.7, and because (G2), with the measures of Definition 2.1, is exactly the familiar linear normalization $f(1) = f_D(1)$. Part (b) is not used elsewhere. Lemma 5.10 also gives an alternative proof of Theorem 5.8(b) near the center, without (L2): subtract $\tilde{f}(1)\tilde{f}^{\mathrm{St}}$, after which the orbital integrals vanish near $\overline{Z}_v$ by Lemma 5.10 and Corollary 5.11.

6 Theorem C′: the global transfer improved

This section deduces Theorem C′ from Theorem E. We relax Part I's notion of an admissible pair of global test functions so that arbitrary matching functions are allowed at all places of S but one (§6.1), check that the comparison of trace formulas and the packet identity of Part I still hold (§6.2), and prove Theorem C′ and its converse (§6.3). Finally we describe exactly when the new non-cancellation hypothesis can fail (§6.4). Notation is that of Part I, §§5–7. The section depends on Theorem E and on Part I, §§5–7, in the form (I7)–(I11). It introduces u-admissible pairs and the hypotheses (H'_D) and (H′).

6.1 u-admissible pairs and the comparison

We define the class of global test functions used in this section (u-admissible pairs, Definition 6.1) and show that Part I's comparison of trace formulas applies to it.

Definition 6.1. Fix a finite place $w_0 \notin S$, a place $u \in S$, and a finite set $S_D \supseteq S \cup \{w_0\}$ as in Part I, §5.3. A pair $(\tilde{f}_D, \tilde{f})$ of factorizable genuine test functions with central characters $\tilde{\theta}^{-1}$ is *u-admissible* if:

(a) at u: $\tilde{f}_{D,u}$ vanishes near $\overline{Z}_u$, $\tilde{f}_u$ is supported on the regular elliptic set, and they match;

(b) at $v \in S \smallsetminus \{u\}$: $\tilde{f}_{D,v}$ and $\tilde{f}_v$ match (no support condition);

(c) for $v \notin S$: conditions (b) and (c) of Part I, Definition 6.5.

By Theorem 5.8(a) and Part I, Theorem 4.5(a), every family $(\tilde{f}_{D,v})_{v \in S}$ with $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$ is the D-side of a u-admissible pair.

Theorem 6.2. ***[P]****For every u-admissible pair,*

$$\sum_{\overline{\pi}_D} m(\overline{\pi}_D) \operatorname{tr} \overline{\pi}_D(\tilde{f}_D) = \sum_{\overline{\pi} \text{ cuspidal}} m(\overline{\pi}) \operatorname{tr} \overline{\pi}(\tilde{f}),$$

both sides converging absolutely.

Proof. We check the proof of Part I, Theorem 6.6, item by item, using the statements (I7)–(I9) of §2.3.

(1) *The D side.* By (I7), with no condition at S, the geometric side is $\operatorname{vol} \cdot \tilde{f}_D(1)$ plus a sum over regular classes; the only non-regular rational elements are central, since r is prime. The term $\tilde{f}_D(1) = \prod_v \tilde{f}_{D,v}(1)$ vanishes because $\tilde{f}_{D,u}$ vanishes near $\overline{Z}_u$.

(2) *The* GL *side.* The hypotheses of (I8) are a supercusp form at w_0 and regular elliptic support at the single finite place u; Definition 6.1(a),(c) provides both, and (I8) imposes nothing at $S \smallsetminus \{u\}$. So the geometric side is a finite sum over regular elliptic rational classes good everywhere, and the spectral side is the cuspidal one.

(3) *Classes whose field does not embed in D.* Such a class is non-elliptic at some $v \in S$ and contributes 0 by (I9), which uses only (M2) at that v; (M2) holds for every matching pair, whatever its support.

(4) *Corresponding classes, signs and measures.* By (I9), which needs only matching at $v \in S$, $F(\iota(\gamma), \tilde{f}_D) = F(\iota(j\gamma), \tilde{f})$; the product of the local signs is 1. The constants $c(\gamma)$ and the global measures are those of Part I, §5.3, which allows any dg_v, $dg_{D,v}$ at $v \in S$; in particular Definition 2.1 (Remark 2.2(1)).

(5) *Places outside S* are as in Part I, Definition 6.5(b),(c).

Summing gives the identity; absolute convergence is that of (I7) and (I8). □

What changes from Part I. Part I, Definition 6.5(a), required regular support at every place of S. Only the place u is needed, for the simple trace formula and to kill the identity term; at the other places of S, Theorem E supplies matching partners of arbitrary functions.

6.2 The packet identity

We extend Part I's packet identity to u-admissible families.

Proposition 6.3. ***[P]****Let ρ, σ, $P_D(\rho)$ and $P(\rho)$ be as in Part I, Proposition 7.6, and let $u \in S$. Let $(\tilde{f}_{D,v}, \tilde{f}_v)_{v\in S}$ be matching pairs with $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$ and $\tilde{f}_u$ supported in the regular set. Then only finitely many terms of each side of*

$$\sum_{\overline{\pi}_D \in P_D(\rho)} m(\overline{\pi}_D) \prod_{v\in S} \operatorname{tr} \overline{\pi}_{D,v}(\tilde{f}_{D,v}) = \sum_{\overline{\pi}\in P(\rho)} m(\overline{\pi}) \prod_{v\in S} \operatorname{tr} \overline{\pi}_v(\tilde{f}_v) \tag{2}$$

are nonzero, and the identity holds.

Proof. The proof of Part I, Proposition 7.6, applies with three changes. (1) The reduction to elliptically supported $\tilde{f}_u$, via Part I, Lemma A.5, is made only at u, where $\tilde{f}_u$ is regular-supported. (2) The uniform finiteness step is Part I, Lemma 7.2 ((I10)), applied at a level $U_S \times U_{w_0} \times U^V$ under which all test functions are bi-invariant. It needs only that every $\tilde{f}_{D,v}$ and $\tilde{f}_v$ ($v \in S$) is bi-invariant under some open compact subgroup over which the cover splits; this holds for every smooth function compactly supported modulo the center, regular-supported or not, and Lemma 7.2 makes no other assumption on the test functions at S. (3) Theorem 6.2 replaces Part I, Theorem 6.6. The separation by archimedean infinitesimal characters and by corner algebras away from $V = S \cup \{w_0\}$ does not involve the test functions at S. □

What changes from Part I. In Part I, Proposition 7.6, all functions at S were regular-supported. Now only the one at u is. In particular, at $v \in S \smallsetminus \{u\}$ one may use the central-character idempotents e_{U_v}, pseudo-coefficients, or supercusp forms (on the GL side, via Theorem 5.8(b)).

6.3 Theorem C′ and its converse

We state the new non-cancellation hypotheses (H'_D) and (H′) (Definition 6.4), which are used again in §6.4 and §7, and prove the transfer in both directions.

Definition 6.4. A packet $P_D(\rho)$ satisfies (H'_D) if there are $u \in S$ and $\tilde{f}_{D,v} \in C_c^\infty(\overline{G}_D(K_v), \tilde{\theta}_v)_{\mathrm{gen}}$ ($v \in S$), with $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$ and the others arbitrary, such that

$$\sum_{\overline{\pi}'_D \in P_D(\rho)} m(\overline{\pi}'_D) \prod_{v\in S} \operatorname{tr} \overline{\pi}'_{D,v}(\tilde{f}_{D,v}) \neq 0.$$

A packet $P(\rho)$ satisfies (H′) if there are $u \in S$ and $\tilde{f}_v \in C_c^\infty(\overline{G}_v, \tilde{\theta}_v)_{\mathrm{gen}}$ ($v \in S$), all with vanishing orbital integrals at good regular non-elliptic elements, with $\tilde{f}_u$ supported in the regular set, such that $\sum_{\overline{\pi}'\in P(\rho)} m(\overline{\pi}') \prod_{v\in S} \operatorname{tr} \overline{\pi}'_v(\tilde{f}_v) \neq 0$.

The test functions allowed in Part I's (H_D) (Definition 7.9) and (H) (Theorem 7.12) are among those allowed here, so (H_D) ⇒ (H'_D) and (H) ⇒ (H′). We do not know whether these implications are strict.

Theorem 6.5 (Theorem C′)**.** ***[P]*** *under* (H'_D) *Let $\overline{\pi}_D$ be an irreducible genuine automorphic representation of $\overline{G}_D(\mathbb{A}_K)$ with central character $\tilde{\theta}$, such that $\overline{\pi}_{D,w_0}$ is supercuspidal for some finite place $w_0 \notin S$. Let $\rho = \overline{\pi}_D^V \circ (\psi^V)^{-1}$ and $\sigma = \overline{\pi}_{D,w_0} \circ \psi_{w_0}^{-1}$, and assume that $P_D(\rho)$ satisfies* (H'_D)*. Then there is a genuine cuspidal automorphic representation $\overline{\pi}$ of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$ such that $\overline{\pi}_v \cong \overline{\pi}_{D,v} \circ \psi_v^{-1}$ for every $v \notin S$, and* (2) *holds for all families as in Proposition 6.3.*

Proof. Take u and $(\tilde{f}_{D,v})_{v\in S}$ as in (H'_D). At u, Part I, Theorem 4.5(a), gives a matching $\tilde{f}_u$ supported on the good regular elliptic set; at $v \neq u$, Theorem 5.8(a) gives matching $\tilde{f}_v$. By Proposition 6.3 the right side of (2) is nonzero, so $P(\rho) \neq \emptyset$. Any $\overline{\pi} \in P(\rho)$ is as required. □

Remark 6.6 (Two roles of u)**.** In Proposition 6.3 the place $u \in S$ is an arbitrary *chosen* test place: the packet identity holds for every choice of u, with the corresponding families. In (H'_D) the place u is *existential*: it suffices that for at least one $u \in S$ some admissible test family gives a nonzero packet sum. Theorem C′ uses that u in Proposition 6.3.

What changes from Part I. Part I, Theorem 7.11, assumed (H_D), which allows only test functions vanishing near the center at *every* place of S. Theorem C′ needs this at one place only. Proposition 6.9 describes what a failure of (H'_D) would require.

Theorem 6.7 (Converse)**.** ***[P]*** *under* (H′) *Let $\overline{\pi}$ be a genuine cuspidal automorphic representation of $\overline{G}(\mathbb{A}_K)$ with central character $\tilde{\theta}$, with $\overline{\pi}_{w_0}$ supercuspidal for some finite $w_0 \notin S$, and let $\rho = \overline{\pi}^V$, $\sigma = \overline{\pi}_{w_0}$. If $P(\rho)$ satisfies* (H′)*, there is $\overline{\pi}_D$ with $\overline{\pi}_{D,v} \cong \overline{\pi}_v \circ \psi_v$ for all $v \notin S$.*

Proof. Take u and $(\tilde{f}_v)_{v\in S}$ as in (H′). At u, Part I, Theorem 4.5(b), gives a matching $\tilde{f}_{D,u}$ vanishing near $\overline{Z}_u$; at $v \neq u$, Theorem 5.8(b) gives matching $\tilde{f}_{D,v}$. Proposition 6.3 applies, and its right side is nonzero. Hence $P_D(\rho) \neq \emptyset$. □

Corollary 6.8. ***[P]*** *(using **[E]**) Suppose $P(\rho) = \{\overline{\pi}\}$ (with any multiplicity) and $\overline{\pi}_v$ is square-integrable for every $v \in S$. Then* (H′) *holds, and $\overline{\pi}$ transfers to $\overline{G}_D(\mathbb{A}_K)$ as in Theorem 6.7.*

Proof. Fix $u \in S$. For $v \in S \smallsetminus \{u\}$ let $\tilde{f}_v$ be a pseudo-coefficient of $\overline{\pi}_v$ [FK, §25.1, Theorem] [**E**]; its orbital integrals vanish at regular non-elliptic elements, and $\operatorname{tr}\overline{\pi}_v(\tilde{f}_v) = 1$. At u, the character $\Theta_{\overline{\pi}_u}$ is not identically zero on the good regular elliptic set, by the orthonormality of characters of square-integrable genuine representations on the elliptic set [FK, §25], from [Kaz, Theorem K] [**E**]. As in Part I, Remark 7.13(1), a bump function $\tilde{f}_u$ supported near a good regular elliptic point where $\Theta_{\overline{\pi}_u} \neq 0$ has $\operatorname{tr}\overline{\pi}_u(\tilde{f}_u) \neq 0$ and vanishing non-elliptic orbital integrals. The sum in (H′) is $m(\overline{\pi}) \prod_v \operatorname{tr}\overline{\pi}_v(\tilde{f}_v) \neq 0$. □

What changes from Part I. Part I, Remark 7.13(1), obtained (H) for a single $\overline{\pi}$ only when its components at S are supercuspidal (or, at each place, have character not identically zero on the elliptic set, using regular-supported functions everywhere). Now square-integrable components suffice, because pseudo-coefficients, which are not regular-supported, can be used at all places of S but one.

6.4 When can (H'_D) fail?

We describe the coefficient functions of packets for which (H'_D) fails. This sharpens Part I, Lemma 7.8.

For a packet $P_D(\rho)$ let $m(\tau)$ be the total multiplicity of its members whose component at S is $\tau = \otimes_{v\in S}\tau_v$, a genuine irreducible representation of $\prod_{v\in S}\overline{G}_D(K_v)$ with central character $\tilde{\theta}_S$.

Proposition 6.9. ***[P]****(H'_D) fails for $P_D(\rho)$ if and only if there is $c \geq 0$ with $m(\tau) = c\prod_{v\in S}\dim\tau_v$ for every τ. Here $c = 0$ exactly when $P_D(\rho)$ is empty. The constant c need not be an integer.*

Proof. Step 1 (fixed level). Call a family $U = (U_v)_{v\in S}$ of open compact subgroups a *level* if each U_v is as in Part I, Lemma 7.8 (split by κ_v, with $\tilde{\theta}_v \circ \kappa_v$ trivial on $U_v \cap K_v^\times$). Let $\mathcal{H}_v$, $\mathcal{R}_v$, $\mathcal{T}_v$ be as there. Levels form a directed family (any two contain a common level) that is cofinal among open compact subgroups, and every $\tilde{f}_{D,v}$ lies in $\mathcal{H}_v$ for some level, since it is bi-invariant under small enough split subgroups. At a fixed level, $\operatorname{tr}\overline{\pi}'_{D,v}(\tilde{f}_{D,v}) = 0$ unless $\overline{\pi}'_{D,v} \in \mathcal{T}_v$. The members of $P_D(\rho)$ with a given component τ at S have all their components fixed, so their total multiplicity $m(\tau)$ is finite. Hence on $\bigotimes_v \mathcal{H}_v$ the packet sum of Definition 6.4 equals

$$\Lambda_U\Big(\textstyle\bigotimes_v f_v\Big) = \sum_{\tau\in\prod_v \mathcal{T}_v} m(\tau)\prod_{v\in S}\operatorname{tr}\tau_v(f_v),$$

a finite sum. So (H'_D) fails if and only if, for every level U and every $u \in S$, Λ_U vanishes on $\mathcal{R}_u \otimes \bigotimes_{v\neq u}\mathcal{H}_v$.

Step 2 (coordinates). By the proof of Part I, Lemma 7.8, $\mathcal{H}_v \cong \bigoplus_{\tau_v\in\mathcal{T}_v}\operatorname{End}(\tau_v^{U_v})$, so the map $\mathcal{H}_v \to E_v := \mathbb{C}^{\mathcal{T}_v}$, $f \mapsto (\operatorname{tr}\tau_v(f))_{\tau_v}$, is surjective; and $\mathrm{ev}_{1,v} = \operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v)^{-1}\sum_{\tau_v}(\dim\tau_v)\operatorname{tr}\tau_v$ on $\mathcal{H}_v$, with $\mathcal{R}_v = \ker\mathrm{ev}_{1,v}$. So the image of $\mathcal{R}_v$ in E_v is the hyperplane $d_v^\perp$, where $d_v = (\dim\tau_v)_{\tau_v}$ and $\perp$ is taken for the bilinear pairing $\langle x, y\rangle = \sum x_\tau y_\tau$. Write m_U for the restriction of m to $\prod_v\mathcal{T}_v$, an element of $E := \bigotimes_v E_v$; then $\Lambda_U(\otimes f_v) = \langle m_U, \otimes_v(\operatorname{tr}\tau_v(f_v))_{\tau_v}\rangle$. Hence Λ_U vanishes on $\mathcal{R}_u \otimes \bigotimes_{v\neq u}\mathcal{H}_v$ if and only if $m_U \in \mathbb{C}d_u \otimes \bigotimes_{v\neq u}E_v$. If every $\mathcal{T}_v$ is nonempty, so that $d_v \neq 0$, these conditions for all $u \in S$ together say that $m_U \in \mathbb{C}\bigotimes_v d_v$ (use a basis of each E_v whose first vector is d_v). That is, $m(\tau) = c_U\prod_v\dim\tau_v$ for $\tau \in \prod_v\mathcal{T}_v$, and $c_U \geq 0$ since $m \geq 0$. If some $\mathcal{T}_v$ is empty, there is no condition at that level.

Step 3 (independence of the level). If $U' \subseteq U$ componentwise, then $\mathcal{T}_v(U) \subseteq \mathcal{T}_v(U')$, and both identities hold on $\prod_v\mathcal{T}_v(U)$; since all $\dim\tau_v > 0$, $c_{U'} = c_U$ whenever $\prod_v\mathcal{T}_v(U)$ is nonempty. Any two levels contain a common smaller level, and every τ has nonzero fixed vectors at some level. So there is one $c \geq 0$ with $m(\tau) = c\prod_v\dim\tau_v$ for all τ.

Step 4 (converse and the constant). If $m = c\prod_v\dim\tau_v$, then at every level $\Lambda_U(\otimes f_v) = c\prod_v\operatorname{vol}(\overline{G}_D(K_v)/\overline{Z}_v)f_v(1)$, which vanishes when $f_u \in \mathcal{R}_u$; so (H'_D) fails. If $P_D(\rho)$ is nonempty, some $m(\tau) \geq 1$, so $c = m(\tau)/\prod_v\dim\tau_v \in \mathbb{Q}_{>0}$; if it is empty, $m = 0$ and $c = 0$. □

Remark 6.10. The linear algebra in the proof does not use $|S| \geq 2$; when $|S| = 1$, (H'_D) coincides with (H_D), and the proposition sharpens Part I, Lemma 7.8, in that case too.

The weights are $\dim \tau_v$, not $\dim \tau_v^{U_v}$: it is the dimension of the whole representation that occurs in Fourier inversion at the identity. A proof of Proposition 6.9 along the same lines was written independently by ChatGPT (see the Acknowledgments).

Thus a failure of (H'_D) would mean that the packet contains members with *every* possible component at S, with multiplicities proportional to the Plancherel weights. In Part I, failure of (H_D) allowed any coefficient function in the larger space $\sum_{v\in S} d_v \otimes \mathbb{C}^{\prod_{w\neq v}\mathcal{T}_w}$. Removing this last possibility is goal (iv) of Part III. The following consequence reduces it to a finiteness statement.

Corollary 6.11. ***[P]****If $P_D(\rho)$ is nonempty and its members have only finitely many distinct components at S, then (H'_D) holds. The same conclusion holds if the levels at S of the members of $P_D(\rho)$ are bounded, that is, if there is a level U such that every member has nonzero U_v-fixed vectors at each $v \in S$.*

Proof. For each $v \in S$ the group $\overline{G}_D(K_v)/\overline{Z}_v$ is compact and infinite (it contains the infinite profinite group $\mathcal{O}_{D_v}^\times/\mathcal{O}_v^\times$). By the Peter–Weyl theorem for the twisted convolution algebra with central character $\tilde{\theta}_v$, the infinite-dimensional space $L^2(\overline{G}_D(K_v), \tilde{\theta}_v)$ is the Hilbert direct sum of the finite-dimensional spaces $\tau_v \otimes \tau_v^\vee$, so there are infinitely many genuine irreducible τ_v with central character $\tilde{\theta}_v$. If (H'_D) failed, Proposition 6.9 with $c > 0$ (the packet being nonempty) would give $m(\tau) \geq c \prod_v \dim \tau_v > 0$ for every one of the infinitely many $\tau = \otimes_v \tau_v$, so the packet would have infinitely many distinct components at S. For the second statement: only finitely many τ have nonzero U-fixed vectors (Part I, Lemma 7.8: each $\mathcal{T}_v$ is finite). □

Remark 6.12. Positivity alone cannot rule out Plancherel-proportional multiplicities: the regular representation $L^2(H) \simeq \widehat{\bigoplus}_\tau (\dim \tau)\, \tau$ of a compact group H has exactly this shape. Corollary 6.11 shows that what is needed is a bound on the levels at S in a fixed near-equivalence class. This observation is due to ChatGPT (see the Acknowledgments).

7 What is left after Part II

This section lists the goals of Part I, §1.3, that are not achieved here, with what we now know about each, including what Part III (Draft 9) has since proved. Unless stated otherwise, all are **[C]**. The numbering (ii)–(v) follows Part I, §1.3, which refers to it; the section introduces no new notation.

(ii) *Local character relations.* A metaplectic local Jacquet–Langlands correspondence: a bijection $\overline{\pi}_v \leftrightarrow \overline{\pi}_{D,v}$ between genuine square-integrable representations of $\overline{G}_v$ and genuine irreducible representations of $\overline{G}_D(K_v)$, with $\Theta_{\overline{\pi}_{D,v}} = (-1)^{r-1}\Theta_{\overline{\pi}_v} \circ \jmath$ on good regular elements and equal formal degrees for the measures of Definition 2.1. The local inputs are now located: local integrability and character expansions [FK, §13], [Li2, Théorème 4.3.2]; pseudo-coefficients [FK, §25.1]; elliptic orthonormality [FK, §25]; and Theorem E. The main difficulty is globalization: a pseudo-coefficient

isolates a representation only among *tempered* ones, while a global cuspidal spectrum may have non-tempered unitary components at a place of S. A restricted correspondence, for example for supercuspidal $\overline{\pi}_v$, is the natural first target.

Status after Part III. Part III constructs a unitary transform L_D between ordinary invariant class functions on $D_v^\times$ and genuine ones on $\overline{G}_D(K_v)$, with constant $b_D = |n|_{K_v}^{-(r-1)/2}$, and proves that it bijects irreducible characters, preserving dimensions, when the residue characteristic of v is prime to n (Part III, Theorem 7.6). Zou's constant in the Flicker–Kazhdan relation [Z, (3.2)] equals b_D for the cover with parameter c_v, because $\gcd(n, 2rc_v + r - 1) = n$. Part III, Corollary 7.8 (Draft 9), gives the bijection and the character relation by composing L_D, [DKV], and [Z, Proposition 3.6], *provided* the corrected square-integrable Flicker–Kazhdan correspondence is available for the chosen parameters. A sufficient published hypothesis is $(n, N_r) = 1$, where N_r is the least common multiple of the composite positive integers at most r (with $N_r = 1$ if none occur). Part III now checks the cocycle, star map, central-character and root-sum conventions against Part I, Remark 8.1. The result is **[P]** in this stated tame range using the cited **[E]** inputs, not for arbitrary (n, r) merely because $v \nmid n$. At places dividing n, irreducible-character integrality of L_D remains open. The formal-degree statement is consistent with these results. For the measures of Definition 2.1, integrality gives $d(\overline{\pi}_{D,v}) = c_D\, d(\pi_v)$ with $c_D = |n|_{K_v}^{-(r^2-1)/2}$ ($c_D = 1$ at tame places). Zou's relation near the identity, together with Corollaries 3.7 and 5.11, gives $d(\overline{\pi}_v) = c_D\, d(\pi_v)$, provided one has the genuine analog of the relation between the leading germ and the formal degree. Here π_v is the common linear partner.

(iii) *An unconditional global correspondence.* Remove the auxiliary place w_0, either with W.-W. Li's invariant trace formula for covers [Li4] (which assumes an invariant Paley–Wiener theorem at the archimedean places; here these are complex and the covers split there), or by placing a supercusp form at a place of S, which Theorem E(b) now permits on the matching side but which requires (ii) on the spectral side. Then refine the packet identity to individual representations. Multiplicity one and rigidity for $\overline{G}(\mathbb{A}_K)$ are known only for representations lifting to cuspidal representations of $\mathrm{GL}(r, \mathbb{A}_K)$ with elliptic components at three places [FK, Theorem 28]; the argument should not assume more.

(iv) *Removing* (H'_D) *and* (H'). By Proposition 6.9, it suffices to rule out packets whose S-multiplicities are proportional to the Plancherel weights, and by Corollary 6.11 it suffices to bound the level at S of the members of $P_D(\rho)$. One route would be a finite-to-one lifting of $P_D(\rho)$ to automorphic representations of $D_{\mathbb{A}}^\times$, and then to $\mathrm{GL}(r, \mathbb{A}_K)$, where strong multiplicity one gives rigidity; another is to use (ii). *Status:* open.

(v) *Completing [JC, Remark 10.3].* Compose the lifting of [JC] with the transfer, and compare with the Flicker–Kazhdan correspondence, using the corrections of [Z] to [FK, §26.2] and [FK, Lemma 27.1]. *Status:* open.

8 Notation and conventions

This section collects the notation and conventions of the series. Subsections 8.1–8.10 are the shared glossary of the series (the file `glossary.tex` of the project repository, also Part I, §10); they include the status labels (§8.1) and the cocycle normalization (§8.5). Pointers there to Part II refer to the present paper. Subsection 8.11 lists the notation specific to Part II. The section proves nothing and depends on nothing.

8.1 Status labels

Every statement in the series carries one of the following labels. A label is never upgraded without a proof, and every change of label is recorded in the revision log of the draft in which it is made.

[P] *Proved here*: a complete argument is included in the Part in which the label appears, possibly using results labeled **[E]**. A statement proved under an explicitly stated hypothesis is labeled **[P]** together with the hypothesis, as in "**[P]** under (H_D)", and the hypothesis itself is listed separately with its own label.

[E] *Established elsewhere*: a precisely cited theorem from the literature or from an earlier Part, used as a black box. "**[P]** (using **[E]**)" marks a proof that rests on such inputs; "**[E]**/**[P]**" marks a statement that is partly cited and partly proved.

[O] *Proof outlined*: the steps are identified, but at least one of them remains to be written out or checked.

[C] *Conjectural*: new mathematical input is required. This includes the goals deferred to later Parts and the hypotheses under which conditional statements are proved.

[P^{c}] *Proved modulo an exhaustive exact computation*: the proof consists of hand arguments together with a computation that is exhaustive over an explicitly described finite set and uses exact arithmetic only (no sampling, and no floating point without exact re-verification). Each such script is committed to the repository, cited by commit hash, and described in an appendix and in the scripts' README. At present this label is used only in the companion paper on the rank-two wild place.

8.2 Global data

K, n, μ_n K is a totally imaginary number field containing the group μ_n of n-th roots of unity, and $n \geq 1$ is odd. All archimedean places of K are complex.

r, D, L, S r is a prime with $(n, r) = 1$; D is a central division algebra over K of dimension r^2; $L \subset D$ is a cyclic maximal subfield, so $[L : K] = r$; S is the (finite) set of places at which D ramifies. Since r is prime, every $v \in S$ is finite and D_v is a division algebra; $|S| \geq 2$, and $|S|$ is even if $r = 2$ (Part I, §1.1).

r', c, c_v Integers with $rr' \equiv 1 \pmod{n}$ and $n \mid r - 1 + 2c$; then $c_v := r'c$ satisfies $n \mid r - 1 + 2rc_v$ (Part I, Assumption 1.1). Despite the subscript, c_v does not depend on the place v. The parameter c occurs *before* the relabeling of μ_n described in §8.5, and c_v *after* it.

$\mathbb{A}_K$, L_v $\mathbb{A}_K$ is the adele ring of K (written $\mathbb{A}$ when no confusion is possible), and $L_v := L \otimes_K K_v = \prod_{w|v} L_w$.

G_D The algebraic group $D^\times$ over K, so that $G_D(K_v) = D_v^\times$.

Running example $K = \mathbb{Q}(\zeta_3)$, $n = 3$, $r = 2$, $D = (-1, 4-\zeta_3)_K$, ramified exactly at $S = \{\mathfrak{p}_3, \mathfrak{p}_7\}$ with $\mathfrak{p}_3 = (1-\zeta_3)$ and $\mathfrak{p}_7 = (3+\zeta_3)$; $L = K(i) = \mathbb{Q}(\zeta_{12})$, $c = 1$, $r' = 2$, $c_v = 2$. The place $\mathfrak{p}_3$ divides n and $\mathfrak{p}_7$ does not. The note `notes/mjl_running_example.tex` works the example out across the series.

8.3 Local fields and absolute values

F, $\mathcal{O}_F$, ϖ, q_F, p F is a nonarchimedean local field of characteristic zero containing μ_n (usually $F = K_v$), with ring of integers $\mathcal{O}_F$, uniformizer ϖ, residue field of order q_F, and residual characteristic p.

$|\cdot|_F$, $|\cdot|_p$ $|\cdot|_F$ is the normalized absolute value, $|\varpi|_F = q_F^{-1}$; on $\mathbb{Q}_p$ it is written $|\cdot|_p$, with $|p|_p = p^{-1}$. In particular $|n|_F = 1$ if $p \nmid n$, and $|F^\times / F^{\times n}| = n^2 |n|_F^{-1}$ (Part I, proof of Proposition 3.5).

Places dividing n; tame and wild places A place v (or the field $F = K_v$) *divides* n if $p \mid n$. The places $v \in S$ with $p \nmid n$ are called *tame*, and those with $p \mid n$ *wild*. In the running example $\mathfrak{p}_7$ is tame and $\mathfrak{p}_3$ is wild ($|3|_{\mathfrak{p}_3} = 3^{-2}$).

8.4 Hilbert symbols

$(a, b)_F$ The n-th Hilbert symbol of F, with values in μ_n; Part IV and the companion paper also write $(a, b)_{n,F}$ or $(a, b)_n$. It is bimultiplicative, nondegenerate and antisymmetric, $(a, b)_F = (b, a)_F^{-1}$. At complex places it is trivial.

Norm compatibility For a finite extension F'/F, $a \in F^\times$ and $b \in F'^\times$, $(a, b)_{F'} = (a, N_{F'/F} b)_F$; in particular $(a, b)_{F'} = (a, b)_F^{[F':F]}$ for $a, b \in F^\times$ [Serre, *Local Fields*, Ch. XIV].

$(a, b)_{\mathbb{A}}$ For a number field $M \supseteq K$, $(a, b)_{\mathbb{A}_M} = \prod_w (a_w, b_w)_{M_w}$. Hilbert reciprocity: $(a, b)_{\mathbb{A}_M} = 1$ for $a, b \in M^\times$.

8.5 Covers and the cocycle normalization

Cover, p, $\overline{H'}$ An *n-fold cover* of a locally compact group H is a topological central extension $1 \to \mu_n \to \overline{H} \xrightarrow{p} H \to 1$. For a subgroup $H' \subset H$ we write $\overline{H'} :=$

$p^{-1}(H')$. Parts I and II write $\overline{H}$; Parts III, IV and the companion paper write $\widetilde{H}$, $\widetilde{Z}$.

Commutators For lifts $\tilde{x}, \tilde{y}$ of commuting elements, $[x, y] := \tilde{x}\tilde{y}\tilde{x}^{-1}\tilde{y}^{-1} \in \mu_n$; it does not depend on the lifts, and it is bimultiplicative on a torus [Flicker–Kazhdan, §3].

Kazhdan–Patterson cocycle σ_F The cocycle of [Kazhdan–Patterson, §0.1] and [Flicker–Kazhdan, §2] on $\mathrm{GL}(r, F)$ with twisting parameter 0. On the diagonal torus, $\sigma_F(\mathrm{diag}(a_i), \mathrm{diag}(b_i)) = \prod_{i<j}(a_i, b_j)_F$.

Parameter-c' cover $\overline{G}^{(c')}$ The cover of $\mathrm{GL}(r, F)$ with cocycle $\sigma_F(g, h)\,(\det g, \det h)_F^{c'}$. The comparison uses $c' = c_v$; the parameter is forced (Part I, Remark 3.3). On the diagonal torus the commutator of $\mathrm{diag}(t_i)$ and $\mathrm{diag}(u_j)$ is $\prod_{i,j}(t_i, u_j)_F^{A_{ij}}$ with $A = -I_r + (1 + 2c_v)J_r$, where J_r is the all-ones matrix (Part IV, §2.1).

Ambient cover $\overline{\mathcal{G}}_v$ The cover of $\mathcal{G}_v = \mathrm{GL}(r, L_v) = \prod_{w|v} \mathrm{GL}(r, L_w)$ with cocycle $\prod_{w|v} \sigma_{L_w}(g_w, h_w)\,(\det g_w, \det h_w)_{L_w}^{c_v}$, the copies of μ_n being identified (Part I, §2.2 and equation (1)). Both covers compared in the series are subgroups of it, so their common central subgroup $p^{-1}(K_v^\times)$ is literally the same subgroup on the two sides.

Restriction and relabeling Pulled back along $\iota_D : D_v^\times \hookrightarrow \mathcal{G}_v$, the ambient cocycle is the local component of the cocycle $\beta = \sigma_L \cdot (\mathrm{Nrd}, \mathrm{Nrd})^c$ of the corrected 1988 paper [JC, equation (4)]. Pulled back along the diagonal embedding of $\mathrm{GL}(r, K_v)$, it is isomorphic, *as an extension*, to the extension with cocycle $\sigma_F^r(\det, \det)_F^c$ (Part I, Proposition 2.3). The group μ_n is relabeled *once*, at every place and globally, by $\zeta \mapsto \zeta^{r'}$; after relabeling the second extension is $\overline{G}^{(c_v)}$. Formulas state whether they hold before or after relabeling. The isomorphism is one of extensions, not an equality of cocycles, and it is unique up to $\tilde{g} \mapsto \chi(\det p(\tilde{g}))\,\tilde{g}$ with $\chi : F^\times \to \mu_n$ continuous. Central characters are therefore never transported through it (Part I, Remark 2.5).

Scalar centrality Because $n \mid r - 1 + 2rc_v$, the whole inverse image $\overline{Z} = p^{-1}(F^\times)$ of the scalars is central (Part I, Lemma 2.2). A cover with this property is called *scalar-central*.

Global normalization Globally one uses the normalized cocycle $\tau_w(g, h) = \sigma_w(g, h)\,\kappa_w(g)\kappa_w(h)\kappa_w(gh)^{-1}$ of [Flicker–Kazhdan, §2], where κ_w is trivial for $w \mid n\infty$. The compact splitting is $\kappa_w^*(k) = (k, 1)_\tau$ on $\mathrm{GL}(r, \mathcal{O}_{L_w})$ for $w \nmid n\infty$, and the rational splitting is $\iota(x) = s_\tau(x)\,\kappa_L(x)^{-1}$; in the unnormalized models the rational splitting is $x \mapsto (x, 1)_\sigma$ at every place (Part I, Lemma 5.1). The global ambient cover $\overline{\mathcal{G}}(\mathbb{A}_L)$ has cocycle $\tau_L \cdot (\det, \det)_{\mathbb{A}_L}^{c_v}$.

Star map For h in the base group and any lift $\tilde{h}$, $h^* := \tilde{h}^{\,n}$. It does not depend on the lift, it is conjugation equivariant, $(ah)^* = a^*h^*$ for scalars a, and it is multiplicative on every torus because n is odd (Part III, Definition 2.1; Part IV, §2.2). For $\zeta \in \mu_n(F)$ one has $\zeta^* = 1$.

8.6 Genuine objects, central characters and good elements

Genuine representation A representation of $\overline{H}$ on which μ_n acts by the tautological character $\zeta \mapsto \zeta$.

$\tilde{\theta}$, θ A genuine character $\tilde{\theta}$ of $\overline{Z}$ (unitary from Part I, §5.1 on; globally, a character of $\overline{Z}(\mathbb{A}_K)$ trivial on $\iota(K^\times)$). The same $\tilde{\theta}$ serves both covers, since $\overline{Z}$ is a common subgroup. The ordinary character attached to it is $\theta(a) := \tilde{\theta}(a^*)$; it is trivial on μ_n, and θ does not determine $\tilde{\theta}$ (Part IV, §2.2). Part III requires only that $\tilde{\theta}|_{\mu_n}$ be faithful.

Genuine test functions $C_c^\infty(\overline{H}, \tilde{\theta})_{\mathrm{gen}}$ is the space of smooth functions with support compact modulo $\overline{Z}$ and $f(z\tilde{h}) = \tilde{\theta}(z)^{-1} f(\tilde{h})$ for $z \in \overline{Z}$; in particular $f(\zeta\tilde{h}) = \zeta^{-1} f(\tilde{h})$. Characters and invariant class functions of genuine representations transform by $\tilde{\theta}$ itself (Part III, §2).

Regular, elliptic An element of $\mathrm{GL}(r, F)$ is *regular* if its eigenvalues are distinct, and *elliptic regular* if moreover its characteristic polynomial is irreducible. Since r is prime, the regular elements of $D_v^\times$ are the noncentral ones, and they are all elliptic. An element of a cover is regular (elliptic) if its image is.

Good element A regular $\tilde{\gamma}$ is *good* if it commutes with $p^{-1}(T_\gamma)$, where T_γ is the centralizer of $\gamma = p(\tilde{\gamma})$; equivalently, $\tilde{\gamma}$ is central in the covered torus $\overline{T_\gamma}$. Genuine invariant functions vanish at regular elements that are not good. For a maximal subfield E of D_v (or an elliptic torus $E^\times$ of $\mathrm{GL}(r, F)$), the good elements of $\overline{E^\times}$ are $p^{-1}(F^\times E^{\times n}) = \overline{Z} \cdot \{t^*\}$, of index $|n|_F^{-(r-1)}$, and every genuine irreducible representation of $\overline{E^\times}$ has dimension $|n|_F^{-(r-1)/2}$ (Part I, Proposition 3.5). Covered elliptic tori are abelian exactly at the places not dividing n.

8.7 Orbital integrals and matching

Δ For regular semisimple γ, $\Delta(\gamma) = |D(\gamma)|_F^{1/2} |\det \gamma|_F^{-(r-1)/2}$, where $D(\gamma)$ is the discriminant of the (reduced) characteristic polynomial and det = Nrd on the division-algebra side. Thus $\Delta(\gamma)^2$ is the Weyl discriminant $|\det(1 - \mathrm{Ad}(\gamma^{-1}))|_{\mathfrak{g}/\mathfrak{t}}|_F$ (Part I, Definition 4.1 and Lemma A.5).

Φ, $F = \Delta\Phi$ $\Phi(\tilde{\gamma}, \tilde{f})$ is the orbital integral over $T_\gamma \backslash G$ of a genuine test function at a good regular $\tilde{\gamma}$, and $F(\tilde{\gamma}, \tilde{f}) = \Delta(\gamma)\Phi(\tilde{\gamma}, \tilde{f})$ is the normalized orbital integral, set to 0 at regular elements that are not good (Part I, Definition 4.1). Part III writes $J(t, f) = \Delta(t)\, O(t, f)$ for the same function, with its own measure conventions.

Canonical identification $\jmath$ At $v \in S$ and for a maximal subfield $E \subset D_v$ with an embedding $j : E \hookrightarrow M_r(F)$, every isomorphism of covered tori that is the identity on $\overline{Z}$ has the same restriction $\jmath_{E,j}$ to the good elements; these restrictions induce a bijection $\jmath$ from the good regular classes of the cover of $D_v^\times$ to the good regular elliptic classes of the cover of $\mathrm{GL}(r, F)$ (Part I, Theorem 3.9 and Corollary 3.12).

Matching Genuine test functions $\tilde{f}_D$ and $\tilde{f}$ *match* if (M1) $F(\tilde{\gamma}, \tilde{f}_D) = (-1)^{r-1} F(\jmath(\tilde{\gamma}), \tilde{f})$ at every good regular $\tilde{\gamma}$, and (M2) $F(\tilde{\delta}, \tilde{f}) = 0$ at every good regular non-elliptic $\tilde{\delta}$ (Part I, Definition 4.3). The sign sits in the matching of functions, so that the expected character relation is $\Theta_{\overline{\pi}_D} = (-1)^{r-1} \Theta_{\overline{\pi}} \circ \jmath$.

Supercusp form A genuine test function on the cover of $\mathrm{GL}(r, F)$ whose integrals along the unipotent radicals of all proper parabolic subgroups vanish, for example a matrix coefficient of a supercuspidal representation [Flicker–Kazhdan, §13].

Germ The asymptotic behavior of orbital integrals near a singular element, expressed through Shalika germs [Flicker–Kazhdan, §8]; used near the center in Part II.

8.8 Haar measures

Part I At $v \in S$ the Haar measures on $D_v^\times / F^\times$ and $\mathrm{GL}(r, F)/F^\times$ are arbitrary but fixed, and matching is defined with respect to them; torus measures are transported by conjugation and do not affect matching. At $v \notin S$ the measures correspond under the identifications ψ_v (Part I, §§4.1, 5.3).

Part II At $v \in S$, $\mathrm{vol}(D_v^\times / F^\times) = d(\mathrm{St})^{-1}$, where $d(\mathrm{St})$ is the formal degree of the Steinberg representation for the chosen measure on $\mathrm{GL}(r, F)/F^\times$ (Part II, Definition 2.1).

Part III For the transform L_D, the quotients H/Z and T/Z carry their Haar *probability* measures (Part III, §4).

Formal degree For a representation σ that is square integrable modulo the center, the constant $d(\sigma)$ in the Schur orthogonality relations; it depends on the Haar measure.

8.9 The division-side transform

H, $\widetilde{H}$, L_D For a local division algebra D of degree r over F, $H = D^\times$ and $\widetilde{H}$ is its cover from Part I. The transform L_D sends θ-equivariant invariant L^2 functions φ on H to genuine $\tilde{\theta}$-equivariant invariant functions on $\widetilde{H}$ by $(L_D\varphi)(z\, t^*) = b_D\, \tilde{\theta}(z)\, \Delta(t)\Delta(t^n)^{-1}\varphi(t)$ at good regular elements, and 0 at the others; it is unitary (Part III, Definition 4.1 and Theorem 4.3).

b_D, c_D $b_D = |n|_F^{-(r-1)/2}$ and $c_D = |n|_F^{-(r^2-1)/2}$ (Part III, §§4–5). Both equal 1 at a tame place; in the running example $b_D = 3$ and $c_D = 27$ at $\mathfrak{p}_3$.

Integrality of L_D The question whether L_D carries irreducible characters to irreducible genuine characters. It holds at tame places (Part III) and is open in general at wild places (Part IV; companion paper).

8.10 Named results and hypotheses

Theorems A, B, C Part I: the comparison of covered elliptic tori, the matching of regular-supported test functions, and the global transfer under (H_D) (with its converse under (H)).

Theorems D, E, C′ Part II: the equivariant section near the center and its compatibility with $\jmath$; the matching of arbitrary genuine test functions; and the global transfer under the weaker hypothesis (H'_D) (with its converse under (H′)).

(H_D), **(H)**, (H'_D), (H′) Non-cancellation hypotheses: the character of a near-equivalence packet at the places of S is not identically zero on the relevant matching test functions (Part I, Definition 7.9 and Theorem 7.12; Part II, Definition 6.4 and §6.3). (H_D) holds when the D-side packet is finite (Part I, Proposition 7.10); sufficient conditions for (H) are in Part I, Remark 7.13; and (H_D) $\Rightarrow$ (H'_D), (H) $\Rightarrow$ (H′) (Part II, §6.3).

(D1)–(D4) The steps left open by Part I: matching of arbitrary functions, local character relations, finiteness of near-equivalence packets, and composition with [JC] (Part I, §8). Their status in the later Parts is recorded in Part I, §1.3.

Targets A, B, C The companion paper's three local problems at the wild place of the running example and beyond (companion paper, §1).

8.11 Notation specific to Part II

G, $\overline{G}$, Z, $\overline{Z}$; P_g, A_g; **eigenvalues**
At a finite place v with $F = K_v$: G is $\mathrm{GL}(r, F)$ or $D_v^\times$, $\overline{G} = p^{-1}(G)$, $Z = F^\times$, $\overline{Z} = p^{-1}(Z)$; P_g is the (reduced) characteristic polynomial of g, $A_g = F[g]$, and the eigenvalues of g are the roots of P_g in $\overline{F}$ (§2.1).

Φ^z
The orbital integral over $Z\backslash G$, $\Phi^z(\gamma, f) = \mathrm{vol}(T_\gamma/Z)\,\Phi(\gamma, f)$ for regular elliptic γ [FK, §6] (§2.1).

$d(\mathrm{St})$, **compatible measures,** λ_v
The formal degree of the Steinberg representation of $\mathrm{GL}(r, F)$; the normalization $\mathrm{vol}(D_v^\times/F^\times) = d(\mathrm{St})^{-1}$ at $v \in S$ (Definition 2.1); and $\lambda_v = d(\mathrm{St})\,\mathrm{vol}(D_v^\times/F^\times)$, which equals 1 for these measures (Remark 2.2).

(I1)–(I11)
The results of Part I used here, restated in the form used (§2.3).

Ω_M, V_M, $V_{D,M}$
The monic degree-r polynomials congruent to $(X-1)^r$ modulo $\mathfrak{p}^M$, and the conjugation-invariant open and closed set of elements of G (resp. $D_v^\times$) whose characteristic polynomial lies in Ω_M (§3.1).

δ_n, B, M_0, M_1
: The constants of Lemmas 3.1 and 3.5: $\delta_n = 1$ if $p \nmid n$ and $q^{-e/(p-1)}$ if $p \mid n$, $B = |n|_F^{-1} q^{e/(p-1)}$, M_0 the level from which the binomial series converges, and $M_1 \geq M_0$ the level from which $\tilde{\theta} \circ \sigma$ is trivial on the relevant scalars.

$g^{1/n}$
: The canonical n-th root $\sum_k \binom{1/n}{k}(g-1)^k \in A_g$ for $g \in V_M$, $M \geq M_0$ (Lemma 3.1).

σ, σ_D, σ_{GL}
: The section $\sigma(g) = \tilde{h}^{\,n}$ with $h = g^{1/n}$ (§3.2, Proposition 3.3).

$\mathcal{T}$, $\mathcal{T}_D$; $C_c^\infty(\overline{G}, \tilde{\theta})_{\mathrm{gen},M}$, $C_c^\infty(G/Z)_M$
: The transport $(\mathcal{T}\tilde{f})(p(z)g) = \tilde{\theta}(z)\tilde{f}(z\sigma(g))$, a bijection between genuine functions supported on $p^{-1}(ZV_M)$ and functions on G/Z supported on ZV_M (Proposition 3.6).

$\Gamma_{\mathcal{O}}$, $u_{\mathcal{O}}$
: Shalika germs of $\mathrm{GL}(r, F)$ for the fixed measures, indexed by the unipotent classes $\mathcal{O}$ (partitions of r), with representatives $u_{\mathcal{O}}$; $\Gamma_{\{1\}}$ is the trivial germ (Corollaries 3.7 and 5.11).

(L1), (L2); f_{St}
: The two linear inputs: the Steinberg pseudo-coefficient f_{St} and its orbital integrals, and linear matching of functions with vanishing non-elliptic orbital integrals (§5.1).

e_D, $\tilde{f}^{\mathrm{St}}$
: The genuine function on $\overline{G}_D(K_v)$ equal to $\tilde{\theta}_v^{-1}$ on $\overline{Z}_v \sigma_D(V_{D,M})$, and the transport of $d(\mathrm{St})^{-1} f_{\mathrm{St}}$ cut off to ZV_M; they match (Proposition 5.5).

$\tilde{f}_{\mathrm{reg}}$
: The part of the matching function of Theorem E supported on the good regular elliptic set (Theorem 5.8).

(Ell), (Non); (G1), (G2)
: The orbital-integral form and the unipotent (germ) form of the condition that a function match $\tilde{f}_D$ near $\overline{Z}_v$ (§5.4, Theorem 5.13).

u-admissible pair
: A pair of global test functions with elliptic support at one place $u \in S$ only and arbitrary matching functions at the other places of S (Definition 6.1).

(H'_D), (H')
: The weakened non-cancellation hypotheses (Definition 6.4).

9 Related work

This section discusses the work related to this Part. The literature of the whole series (the Jacquet–Langlands correspondence, covering groups, metaplectic correspondences,

central extensions of division algebras) is reviewed in Part I, §11; we add only what is specific to Part II. The section proves nothing and introduces no notation.

The local results of Part II concern harmonic analysis near the center, and their inputs are of two kinds. For covers of $\mathrm{GL}(r)$, Kazhdan–Patterson [KP2, §1] and Flicker–Kazhdan [FK, §8] proved the germ expansion and the characterization of orbital integrals of genuine functions (after Vignéras), and [FK] constructed the local metaplectic correspondence for square-integrable representations; Zou [Z] corrected and extended parts of that theory. W.-W. Li developed the local harmonic analysis needed for the trace formula on covers [Li2], including semisimple descent, the Plancherel formula and the density of tempered characters, and proved the invariant trace formula for covers [Li4]. On the linear side we use the Shalika germ expansion [Sha72], Howe's work on the Fourier transform and germs of characters for GL_n [How74], Harish-Chandra's theory of admissible invariant distributions [HC70], [HC78], Rogawski's computation of the trivial germ on elliptic tori via the building [Rog81], van Dijk's formula for induced characters [vD72], and, for parabolic descent of germs, Kivinen–Tsai [KT]. Pseudo-coefficients of square-integrable representations and the density theorem are taken in the form stated in [FK, §§25.1, 27.3], which goes back to Kazhdan [Kaz]; the linear Jacquet–Langlands inputs are from [DKV], [Rog83], [Bad02] and [Bad, Theorem 2.2]. The Plancherel formula and Harish-Chandra's finiteness theorem are used in the form of Waldspurger [Wal03], with explicit formulas for $\mathrm{GL}(n, F)$ and $\mathrm{GL}(m, D)$ in Aubert–Plymen [AP]; and Kottwitz's normalization of measures on inner forms is in [Kot88]. For the classification of tempered representations of $\mathrm{GL}(r, F)$ we use Zelevinsky [Zel80]; general background on smooth representations is in Bernstein–Zelevinsky [BZ] and Bernstein [Ber].

Our matching theorem uses none of the deeper parts of the theory of covers. Its main tool is the section σ of §3, which reduces genuine harmonic analysis near the center of either cover to linear harmonic analysis, compatibly with the identification of elliptic tori of Part I. That covers split over small open subgroups is standard [FK, §2]; what is used here is the stronger statement that the splitting given by n-th powers of n-th roots is equivariant and consistent with $\jmath$ on a full conjugation-invariant neighborhood of the unipotent set, at every finite place including those dividing n.

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